ETAS ISOLAR-B 12.7.0

ISOLAR-B Getting Started Guide

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ETAS ISOLAR-B 12.7.0 - ISOLAR-B Getting Started Guide R01 EN – 03.2025

Contents

1 Introduction

This Getting Started Guide provides you with the information you'll require to install and start using ISOLAR-B (Integrated SOLution for AutosaR – Basic Software). A reference to other useful manuals is also provided.

1.1 Purpose

The purpose of this manual is to describe how to use ISOLAR-B. Its ease of use is illustrated through an example project that demonstrates an inter ECU Communication use case.

ISOLAR-B is the ETAS BSW Configuration Tool. It is based on a single domain model (ARTOP) architecture, and it provides a number of advantages when building AUTOSAR system and ECU software

Abstract: AUTOSAR is an evolving standard, which defines modular software architecture for automotive electronic control units. It is designed to increase the ease of software re-use and the deployment on different hardware. AUTOSAR increases the ease of software integration, but in a way that mandates the use of tools to effectively handle its complexity.

Key features of ISOLAR-B include:

Automatic Generation of BSW

- AUTOSAR Top-Down approach for automatic generation of BSW based on System, ECU-EXTRACT and diagnostic descriptions.
- Supports OEM extensions for special use cases and parameters.

Iterative Development

- Special import features for updating the BSW Values after a change has been made to the input data given by the OEM.
- Advanced Diff/Merge tool to import the delta modified information.
- Rule Based Comparator to hide the non-relevant information.

Enhanced User Experience

- Special GUIs in the BSW for high productivity and faster configuration.
- Message flow view to allow easy debugging and analysis of complex use cases (Gateway, Multiplex).
- Guided Workflows for easy navigation.
- Voice Enabled features.

ISOLAR-B supports both basic and advanced (i.e. variant handling, Post-build etc.) AUTOSAR BSW editing use cases.

Who Should Read this Manual?

This manual will be of interest to anyone who wishes to install and use ISOLAR-B to develop AUTOSAR compatible ECU and/or basic software. You should read this Getting Started Guide before installing ISOLAR-B.

1.2 Document Conventions

`OCI_CANTxMessage msg0`	Code snippets are shown in a monospaced Courier typeface.
Choose **File→Open**.	Menu commands are shown in boldface.
Click **OK**.	Buttons are shown in boldface.
Press <ENTER>.	Keyboard commands are shown in angled brackets.
The 'Open File' dialog box is displayed.	Names of program windows, dialog boxes, fields, etc. are shown in quotation marks.
Select the file *setup.exe*	Text in drop-down lists on the screen, program code, as well as path- and file names are shown in italics.
A *distribution* is always a one-dimensional table of sample points.	General emphasis and new terms are set in italics.

1.3 Privacy Statement (GDPR)

Your privacy is important to ETAS so we have created the following Privacy Statement that informs you which data are processed in ISOLAR-B, which data categories ISOLAR-B uses, and which technical measures you have to take to ensure users' privacy.

Where relevant, we also provide additional information on where ISOLAR-B stores personal or personal-related data, and how it can be deleted.

1.3.1 Data Processing

Personal or personal-related data or data categories are processed if using the ETAS License Manager (LiMa). The purchaser of this product is responsible for the legal conformity of processing the data in accordance with Article 4 No. 7 of the General Data Protection Regulation (GDPR). As the manufacturer, ETAS GmbH is not liable for any mishandling of this data.

When using the ETAS License Manager in combination with user-based licenses, the following personal or personal-related data or data categories are recorded on the license server for the purposes of license management:

- Communication data: IP address
- User data: UserID, WindowsUserID

1.3.2 Technical and organizational measures

This product does not encrypt the personal or personal-related data or data categories that it records. Ensure that the data recorded are secured by means of suitable technical or organizational measures in your IT system. Personal or personal-related data in log files can be deleted by tools in the operating system.

1.4 Safety Notice

WARNING

Any configuration(s) described in this Getting Started Guide are provided as an example only and must be reviewed and adapted for specific project requirements before use.

For any safety related development that uses the ISOLAR-B tool you should apply an applicable safety standard such as ISO 26262:2018.

The ETAS_Safety_Advice.pdf provides additional safety information.

2 Documentation

2.1 Getting Started Guides

- ISOLAR-B Getting Started Guide (This manual)
- ISOLAR-A Getting Started Guide

Note: The example projects used in this manual can be found in the following location:

<ISOLAR-AB_Installed_Location >ExamplesForCustomization08_examples_-for_isolar_b

2.2 Reference Manuals

- ISOLAR-B Help Reference Manual
- ISOLAR-A Help Reference Manual

2.3 Online Help

Detailed help for all the features in ISOLAR-B.

3 System Requirements

Please refer to the latest ISOLAR-B Release Notes for more details.

4 Installation

The Installation and Licensing processes for ISOLAR-B are described in the Release Notes document (Section 2).

5 Working with ISOLAR-B

5.1 Introduction

This section provides you with the basic information you need in order to start working with ISOLAR-B.

5.2 Creating an AUTOSAR Project

5.2.1 Introduction

There are several different ways to create an AUTOSAR Project. The following methods are described in more detail below.

- Create a new AUTOSAR Project from scratch.
- Import an existing AUTOSAR Project Folder.
- Import an existing AUTOSAR Project.
- Import an example AUTOSAR Project.

5.2.2 Create a new AUTOSAR Project from scratch

Use the following steps to create a brand new AUTOSAR project.

1. As shown below, either right-click in the "ECU Navigator" and select **New** › **AUTOSAR Project**, or click the "New Autosar Project" link.

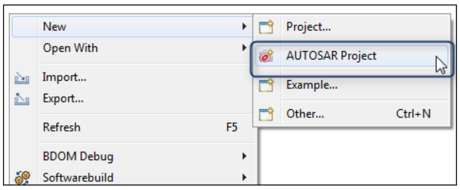

Figure 5.1.Creating a new AUTOSAR Project

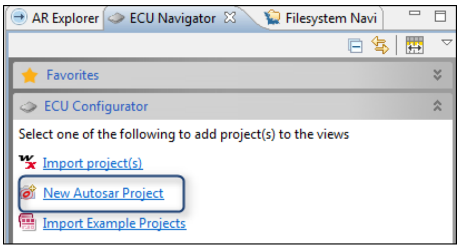

Figure 5.2.Creating a new AUTOSAR Project

2. Whichever method you choose, you'll be taken to the "Create a new AUTOSAR Project" dialog.

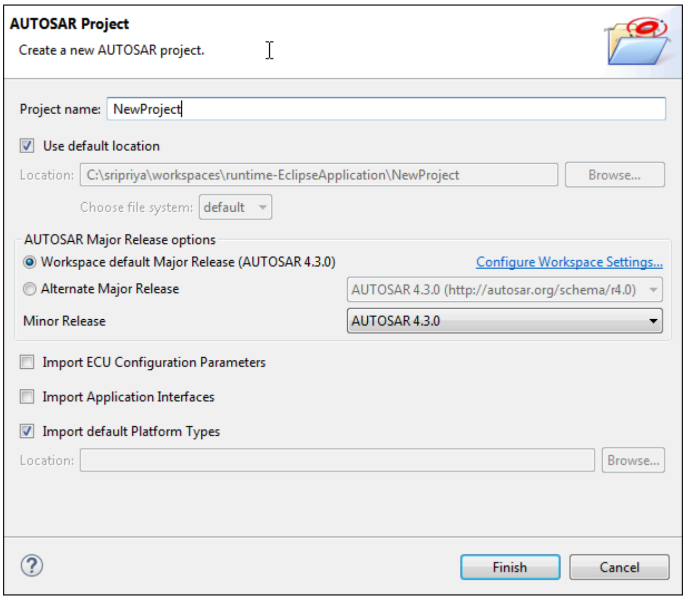

Figure 5.3.Create a new AUTOSAR Project dialog

You will need to provide the following information:

- **Project name:** The name of the new project to be created.
- **Location:** The physical location for the project.
- **Workspace default Major Release:** By default, the latest AR version will be auto-selected.
- **Alternate Major Release:** Use this option to select a different AUTOSAR version for this project.

The following options are also available for controlling the import of existing data into the new project:

- **Import ECU Configuration Parameters:** Import the standard AUTOSAR ECU Parameter Definition file.
- **Import Application Interfaces:** Import the standard Application Interfaces defined by AUTOSAR.
- **Import default Platform Types:** Import the standard Platform Types file defined by AUTOSAR.

3. Click **Finish** to create the new project.

5.2.3 Import an existing AUTOSAR Project Folder

Instead of creating a new project from scratch, if you already have an existing ISOLAR project (that is not currently in your workspace), you can use the folder in which that project is contained as the source of an import. The existing project will be imported (as a new project) into your ISOLAR workspace.

1. Select **File** › **Import...**.

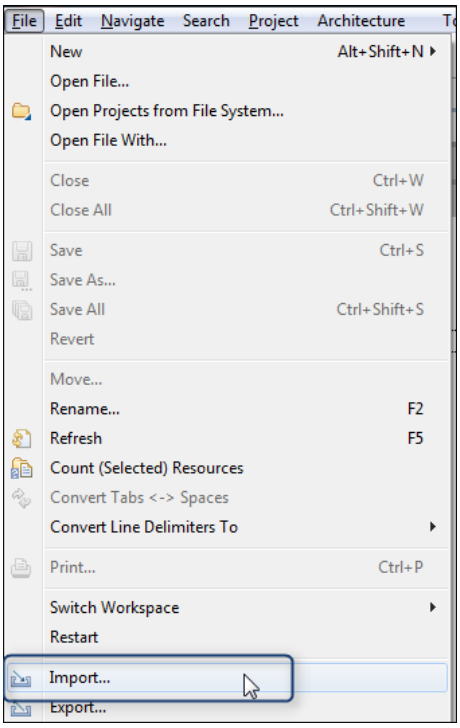

Figure 5.4.Invoking the Import

2. Or, in the "ECU Navigator", right click and select **Import**.

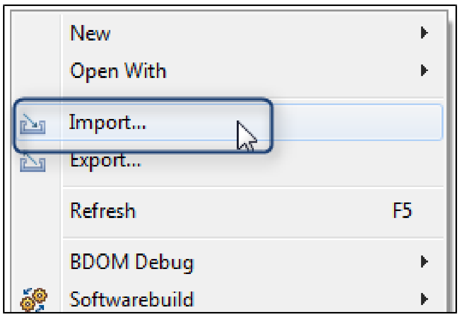

Figure 5.5.Invoking the Import

3. Expand the "General" folder and select the **Import Folder as AR Project** option.

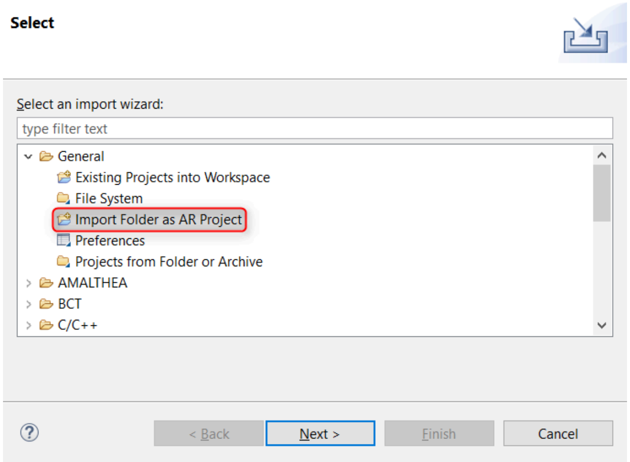

Figure 5.6.Import Folder as AR Project

4. Click “Browse” to locate and select the folder that is to be the source of the import. Note that you can use the “Project Name” field to give the new project a different name from the one that is contained in the source folder.

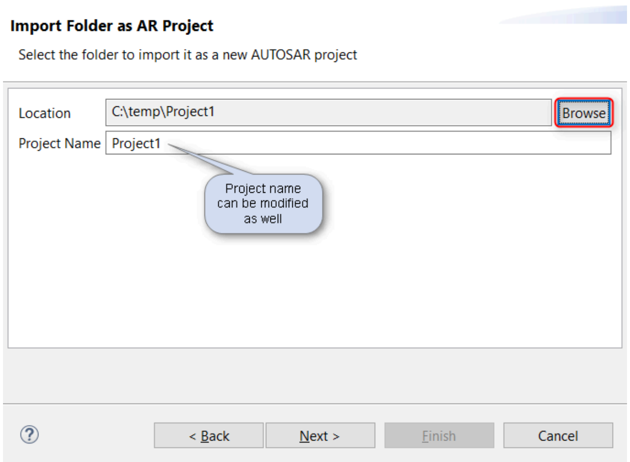

Figure 5.7.Selecting the folder (and naming the new project)

5. Select the AUTOSAR version for this new project (the default will be auto-selected).

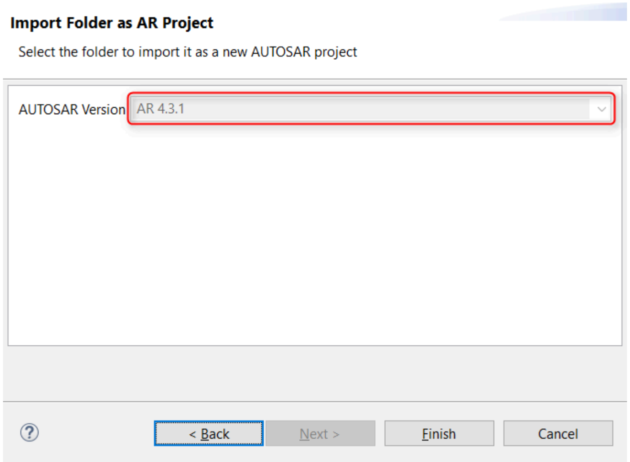

Figure 5.8.Setting the AR version

6. Click **Finish** to import the project contained in the source folder.

5.2.4 Import an existing AUTOSAR Project

It is also possible to import an existing project into your ISOLAR workspace.

1. Invoke the import as shown above, then select the **Existing Projects into Workspace** option.

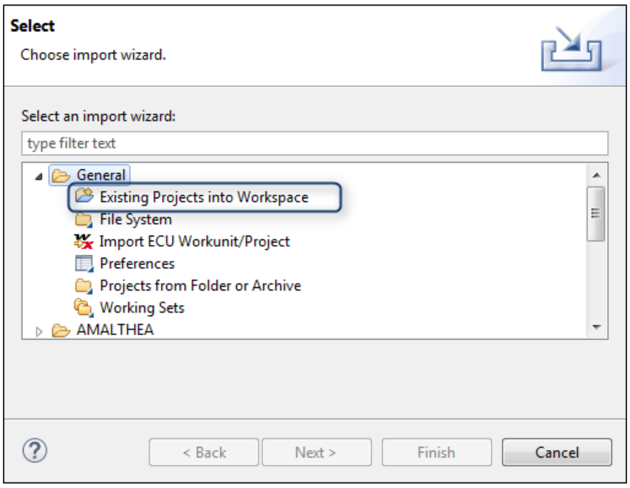

Figure 5.9.Existing Projects into Workspace

2. Click “Browse” to drill down to the location where the existing project is stored, select the project(s) that you wish to import, then click **Finish**.

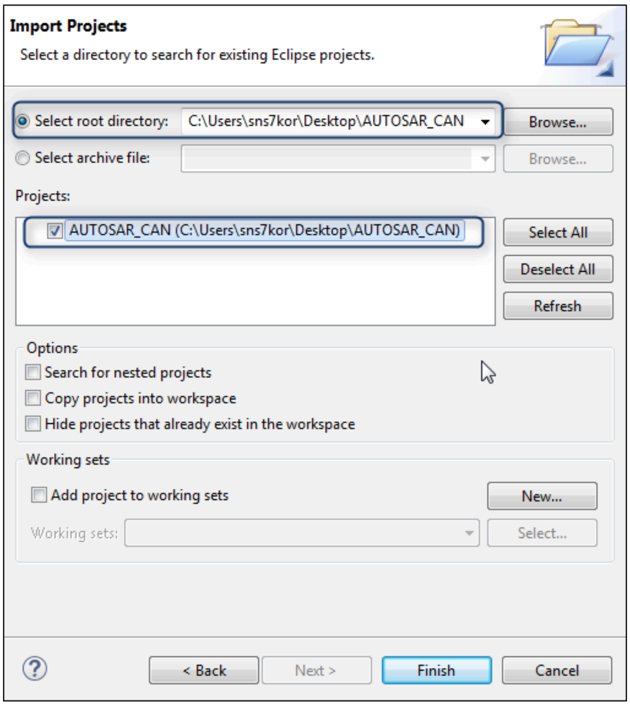

Figure 5.10.Importing a Project

5.2.5 Import an example AUTOSAR Project

A variety of example projects for different usecases is provided with ISOLAR. Each example project contains all the necessary project setup and scripts required for the specific usecase. The following steps explain how to import an example project.

1. As shown below, either right-click in the “ECU Navigator” and select **New** > **Example**, or click the “Import Example Projects” link.

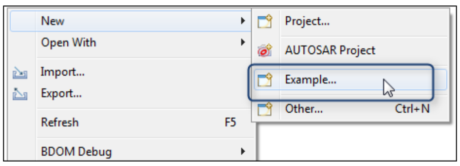

Figure 5.11.Invoking the Example wizard

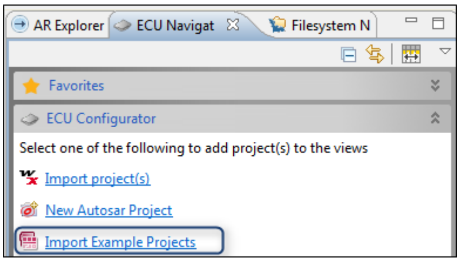

Figure 5.12.Invoking the Example wizard

3. Whichever method you choose, you'll be taken to the "Select a wizard" dialog.

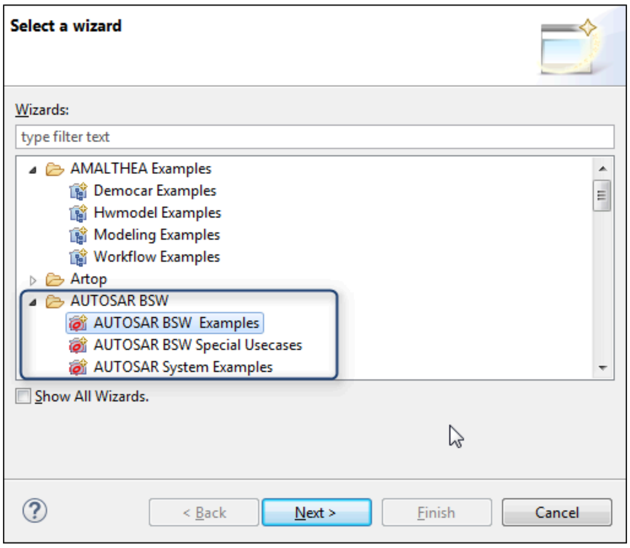

Figure 5.13.Importing AUTOSAR BSW Examples

4. When you open the "AUTOSAR BSW" folder, you'll see that the examples are categorized into 3 types:

- **AUTOSAR BSW Examples:** These examples contain both System Description and BSW configurations, so a Code Generation is immediately possible.
- **AUTOSAR BSW Special Usecases:** These examples contain a sample BSW Configuration for the specific usecase (e.g. NvM).
- **AUTOSAR System Examples:** These examples contain just a System Description, so they can be used to auto-generate BSW configurations ready for Code Generation.

5. Select **AUTOSAR BSW Examples**, then select the specific example that you wish to import.

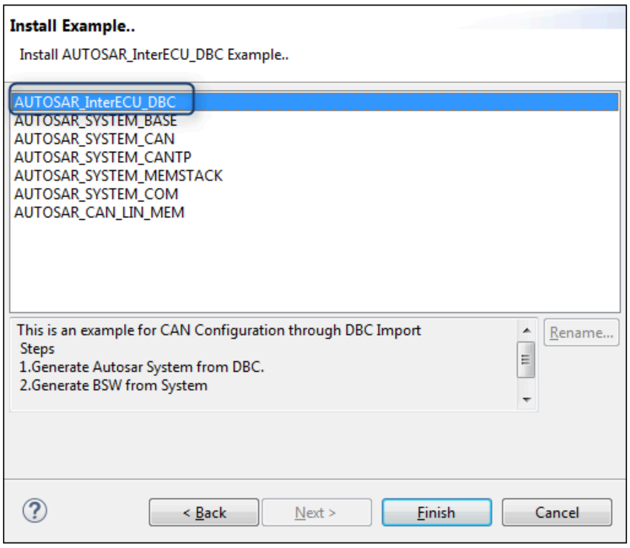

Figure 5.14.Selecting an example

6. Click **Finish** to import the selected example into your workspace.

5.3 Editors and Views

5.3.1 Introduction

This section provides a brief overview of the editors and views available within ISOLAR-B.

5.3.2 ECU Navigator

The ECU Navigator makes it easy to navigate the contents of a project with special consideration for the AUTOSAR-defined BSW stacks and modules.

As shown below, the “ECU Navigator” tab is located on the left side of the AUTOSAR Perspective, and its contents are divided as follows -

- **AUTOSAR template-based grouping:** Software, System, Parameter Definitions, BSW Modules, Bamf, Generic Structure.
- **Stack segregation grouping of the BSW modules** (based on the AUTOSAR defined BSW stack) : E.g. COM Stack, MEM Stack, System Stack, etc.
- **The individual BSW modules are located** (based on the stack to which they belong) : Sub-groups are the stack-level segregation of BSW modules, which are defined in accordance with the AUTOSAR ECU configuration template (e.g. Com, PduR, etc.)

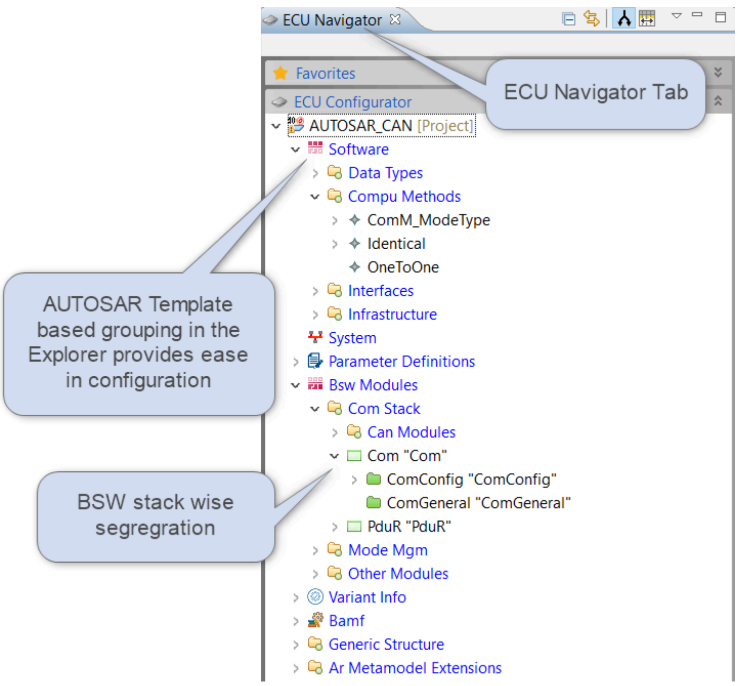

Figure 5.15.ECU Navigator

5.3.3 BSW Editor

The BSW Editor is a generic editor through which all BSW elements can be configured according to paramdef. It can be opened on BSW Elements displayed within the ECU Navigator.

Use either of the two following methods to invoke the editor:

- Double-click a BSW Element under BSW Modules
- Or, as shown here, right click on a BSW Element, then select **Open with › BSW Editor**

Figure 5.16.Invoking the BSW Editor

Here is a summary of the key functionality of the BSW Editor, which is divided into a Tree Section (on the left) for navigation/selection, and the main Editor Section (in the centre).

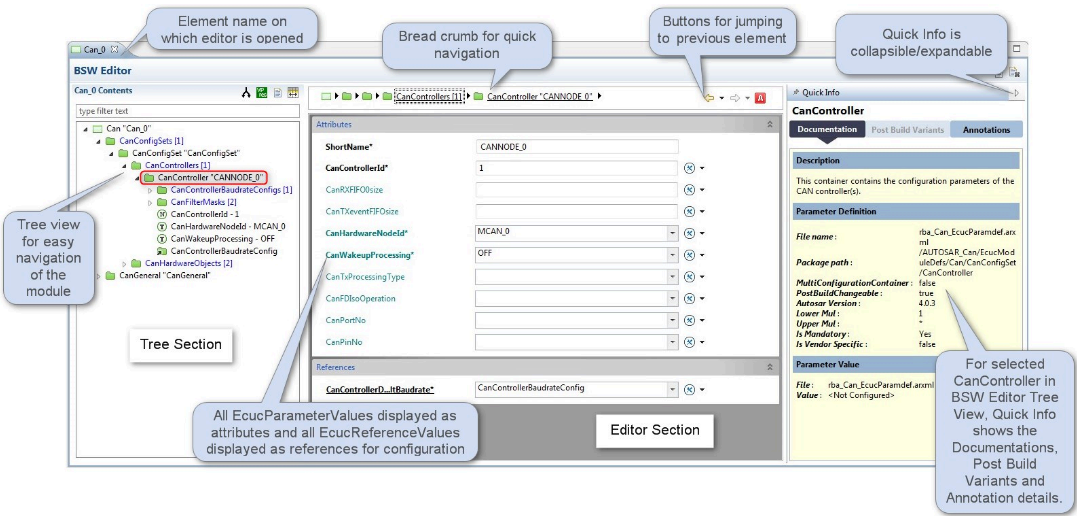

Figure 5.17.BSW Editor Overview

Quick Info section

On the right-hand side of the BSW Editor you'll see the Quick Info section, which can be expanded/collapsed as required.

As you can see below, Quick Info contains tabs for "Documentation", "Post Build Variants" and "Annotations" details for Parameters, Containers and References.

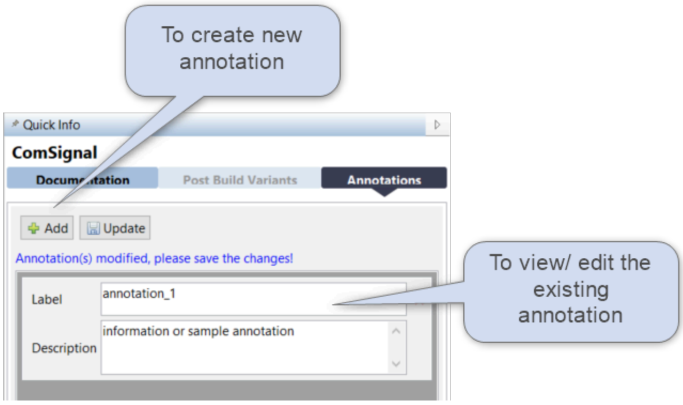

Figure 5.18.Quick Info (Annotations tab)

And here you can see Post Build Variants tab.

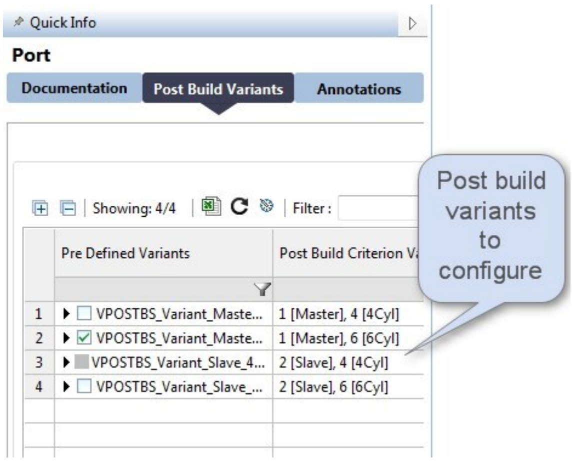

Figure 5.19.Quick Info (Post Build Variants tab)

5.3.4 BswM Abstraction Editor

BswM configurations at different levels can be configured thorough the BswM Abstraction editor, which can be opened on the BswM module displayed within the ECU Navigator.

The editor can be invoked in one of two ways:

1. Double-click a **BswM** module.
2. Or, as shown here, **right click** on the BswM module, then select **Open with** › **BswM Editor**.

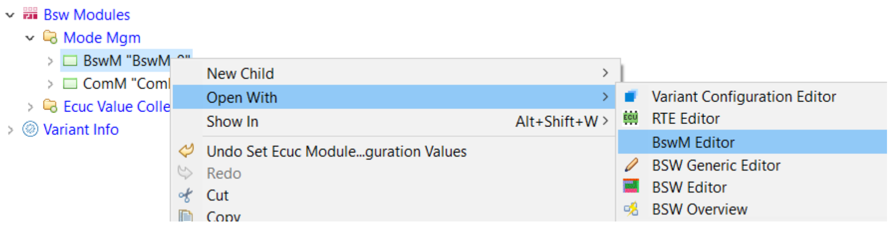

Figure 5.20.Invoking the BswM Editor

Here is a summary of the key functionality of the BswM Abstraction Editor, which is divided into a Rule Section for rule selection (on the left), and IF condition and THEN/ELSE section (on the right).

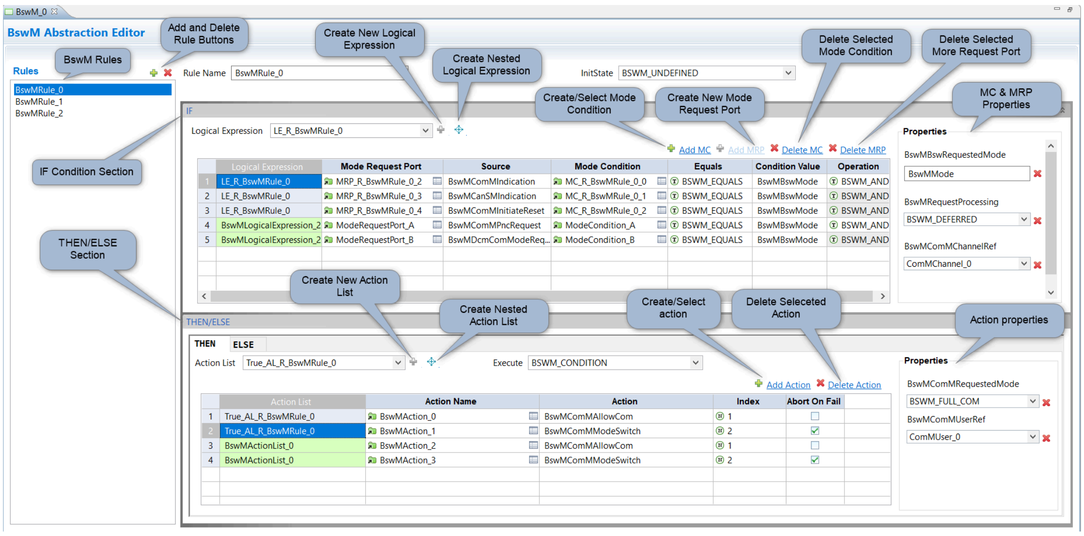

Figure 5.21.The BswM Abstraction Editor

See the ISOLAR-B Reference Guide for more information on the BswM Abstraction Editor.

5.3.5 Os Editor

Os module configurations can be shown in different groups, and these configurations can be updated in the Os Editor, which can be opened on BSW Modules displayed in ECU Navigator.

The editor can be invoked in one of two ways:

1. Double-click the **Os** module.
2. Or, as shown here, **right click** on the Os module, then select **Open with** › **Os Editor**.

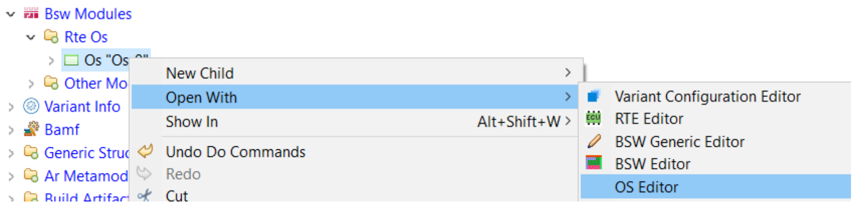

Figure 5.22.Invoking the Os Editor

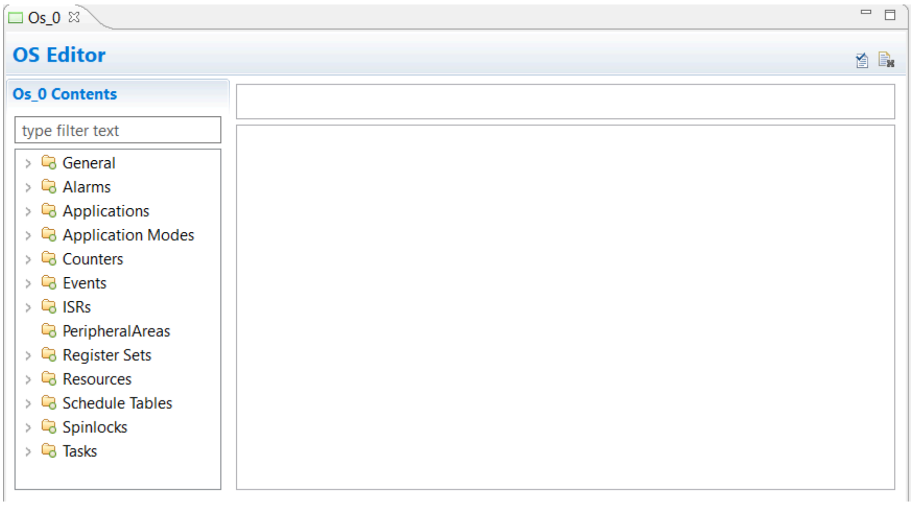

Figure 5.23.The Os Editor

Os Editor Actions

The following actions are available in the Os Editor:

- **New:** Create a container on the selected group (enabled only for groups, which are displayed in yellow).
- **Delete:** Delete a container (enabled only for containers, which are displayed in green).
- **Rename:** Rename a container (enabled only for containers). F2 can also be used for renaming containers.

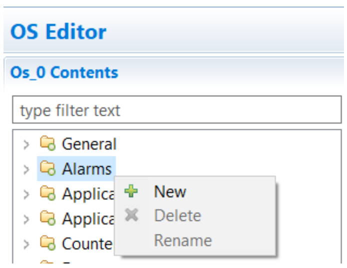

Figure 5.24.Os Editor actions

Os Editor Groups

The following operations can be done on Os Editor groups:

- Clicking a parent container group will show all the subgroups.
- Clicking a subgroup will only show its subgroups.
- All subgroups can be viewed in a single page by selecting the parent container group

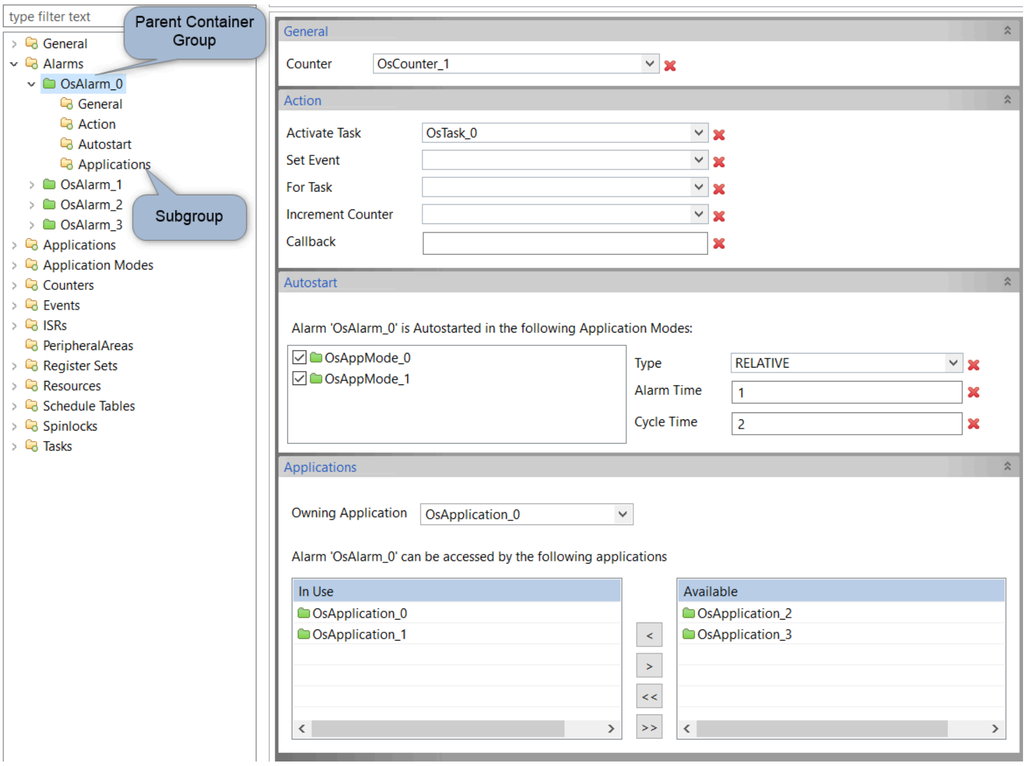

Figure 5.25.Os Editor groups

See the ISOLAR-B Reference Guide for more information on the Os Editor.

5.3.6 AUTOSAR Quick Search

The AUTOSAR Quick Search option allows you to search for BSW elements based on their shortname.

Quick Search can be invoked with a shortcut key (Ctrl + Shift + A) or, as shown here, via the Search menu option.

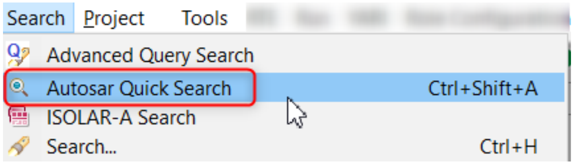

Figure 5.26.Invoking AUTOSAR Quick Search

Once you're in the Autosar Quick Search dialog, just enter the shortname of the BSW element(s) that you're looking for.

The results of your search will be displayed in the Matching Items section, and you can click on any item in the list to open it in the BSW Editor.

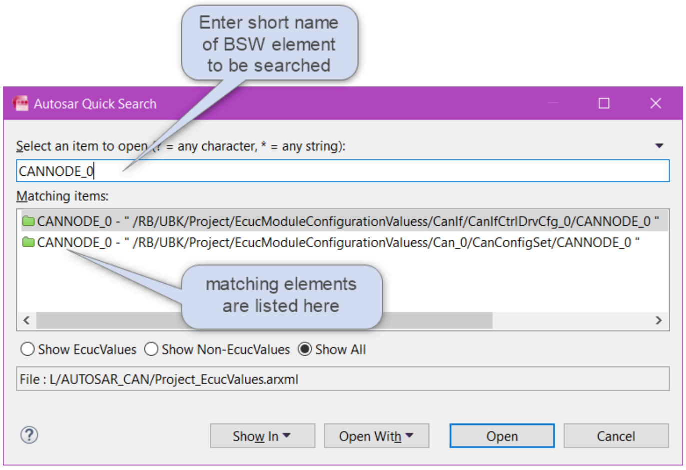

Figure 5.27.AUTOSAR Quick Search

5.4 Query Search

5.4.1 Introduction

The Query Search feature provides search capabilities via a DSL like query language. This allows you to write SQL like queries to search for elements in your project.

5.4.2 Invocation

Query Search is accessed via **Window** › **Show View** › **Other** › **Query Search View**

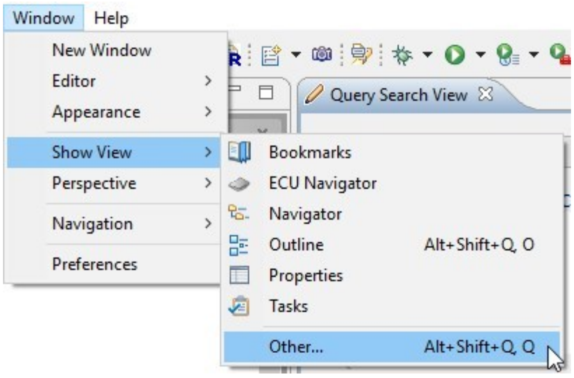

Figure 5.28.Invoking Query Search

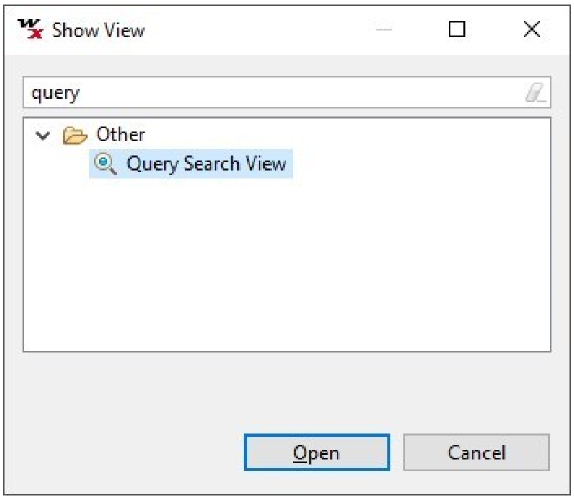

Figure 5.29.Selecting Query Search view

5.4.3 Example

Run a search query as follows:

1. Select the Project in which the search will be made.

2. Enter the query in the text box provided.
3. Click the Search button.

The results of your search will be displayed in the Query Result table.

Here are a couple of sample queries:

```
FROM CanIfCtrlCfg WHERE CanIfCtrlCanCtrlIdRef != null SELECT CanIfCtrlCanCtrlIdRef AS CanIfCtrlCanCtrlIdRef, shortName AS ShortName RETURN CanIfCtrlCanCtrlIdRef
```

```
FROM CanController WHERE CanHardwareNodeId!=null && CanControllerDefaultBaudrate!=null SELECT CanHardwareNodeId AS NodeId
```

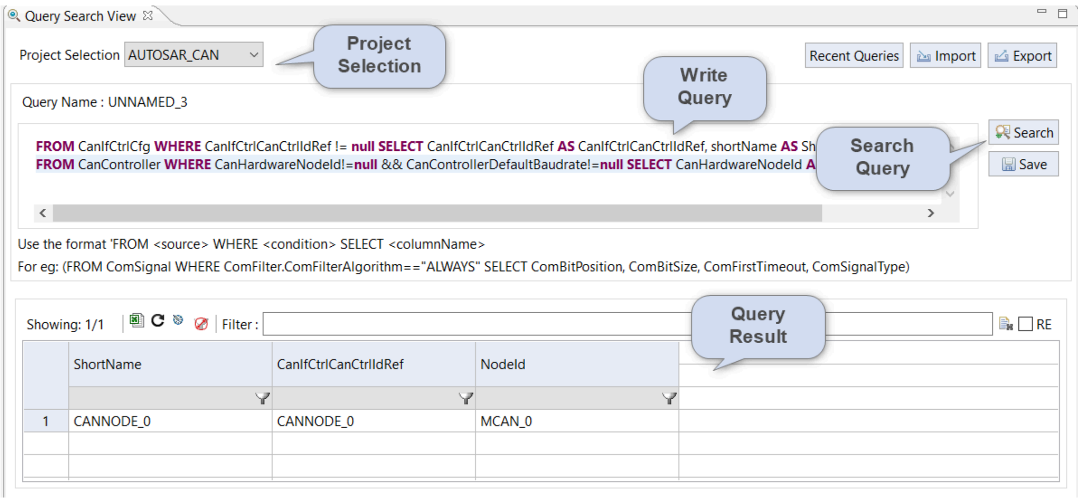

Figure 5.30.Query Search (with result)

5.4.4 Importing/Exporting queries

Queries can be imported/exported.

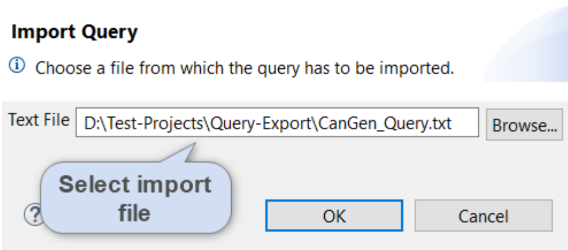

Figure 5.31.Import a query

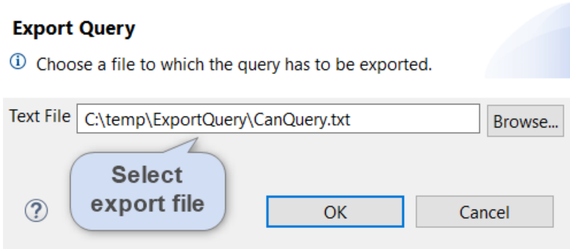

Figure 5.32.Export a query

An entire set of queries, which have previously been run, can be restored using the Recent Queries button.

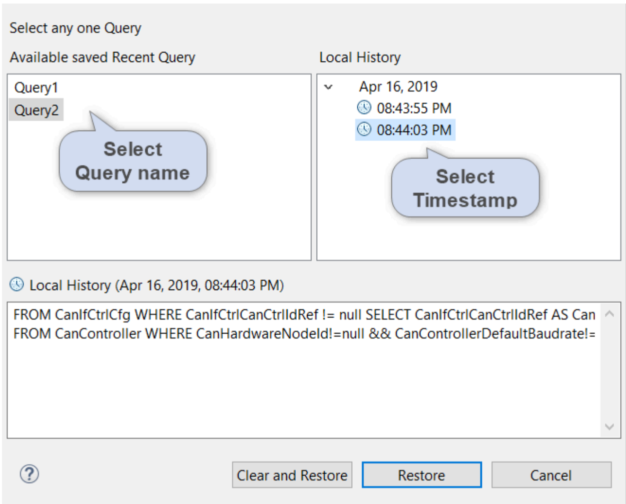

Figure 5.33.Recent queries

5.5 Advanced Query Search

5.5.1 Introduction

The Advanced Query Search feature helps you to generate queries using drag down options in the BSW editor. This allows you to build your queries by selecting from a targeted list of items rather than a complete, non-intuitive list.

5.5.2 Invocation

Advanced Query Search is accessed via **Search** > **Advanced Query Search**.

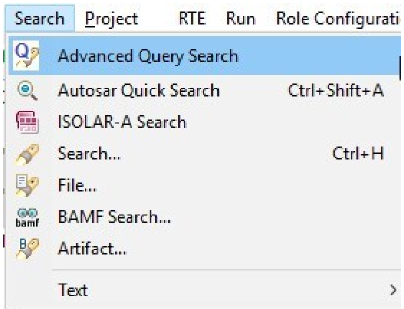

Figure 5.34.Invoking Advanced Query Search

5.5.3 Example

First, select the Project in which you want to run the query, and then elements from the paramdef viewer can be dragged and dropped into the “Source”, “Condition” and “Display” sections to auto-generate the query.

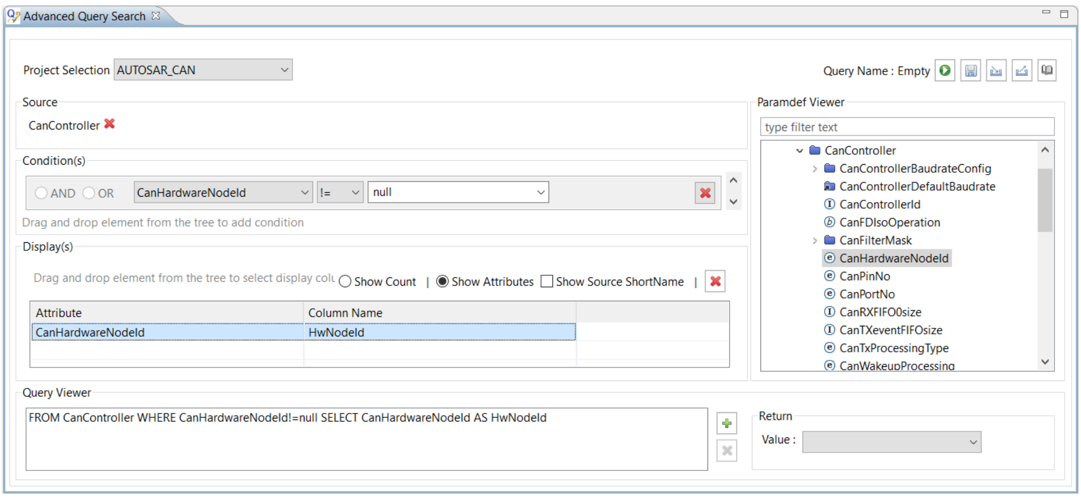

Figure 5.35.Advanced Query Search view

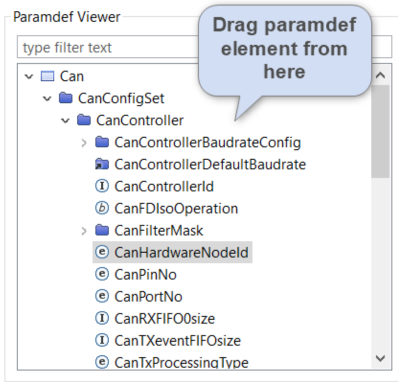

Figure 5.36.Paramdef viewer for Element selection

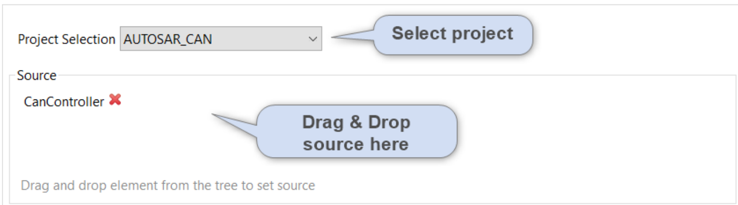

Figure 5.37.Project and Source selection

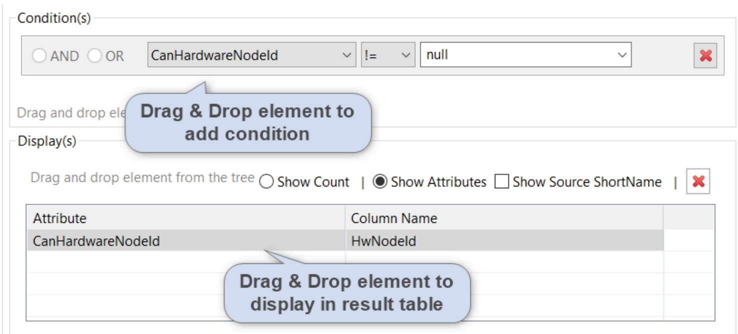

Figure 5.38.Condition and Display selection

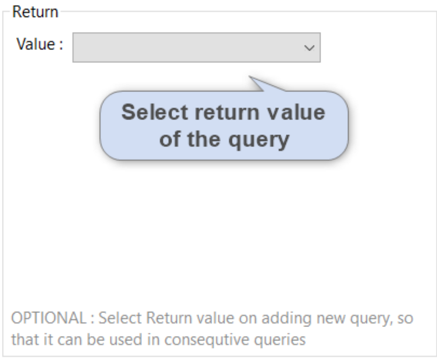

Figure 5.39.Return Value selection

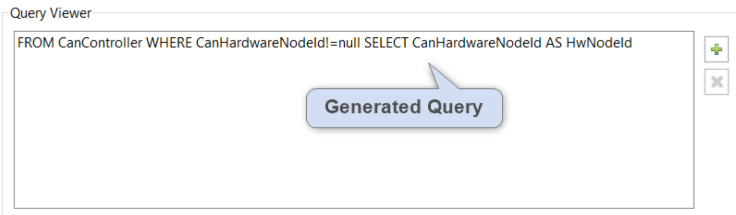

Figure 5.40.Generated Query (in the Query Viewer)

The result of your query will be opened in the Query Result View.

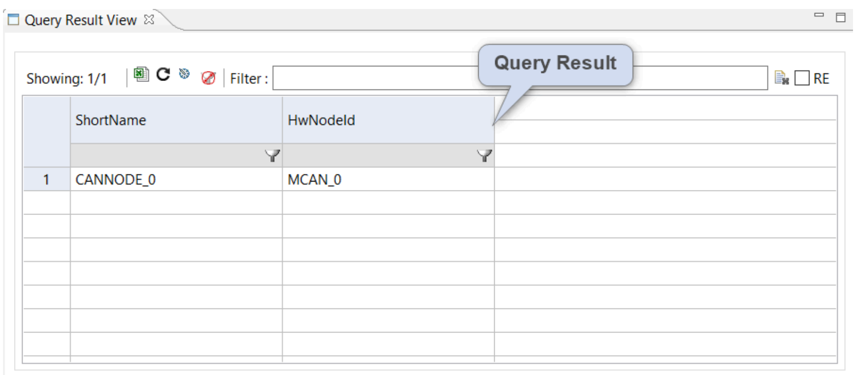

Figure 5.41.Query result

The Import, Export and Recent Queries features, which we described in Section 5.4.4 for Query Search, can also be used in the Advanced Query Search.

5.6 BSW Graphical Views

5.6.1 Introduction

This section describes the various BSW graphical views that are available in ISOLAR-B.

5.6.2 PDU Trace View

The BSW PDU Trace View provides a clear representation of the PDU flow between the layers in the BSW Com stack.

Right-click on any of the following BSW elements in the BSW Editor, and select **PDU Trace View**.

- ComIPdu
- PdurRoutingPath
- CanIfTxPdu/CanIfRxPdu
- CanHardwareObject

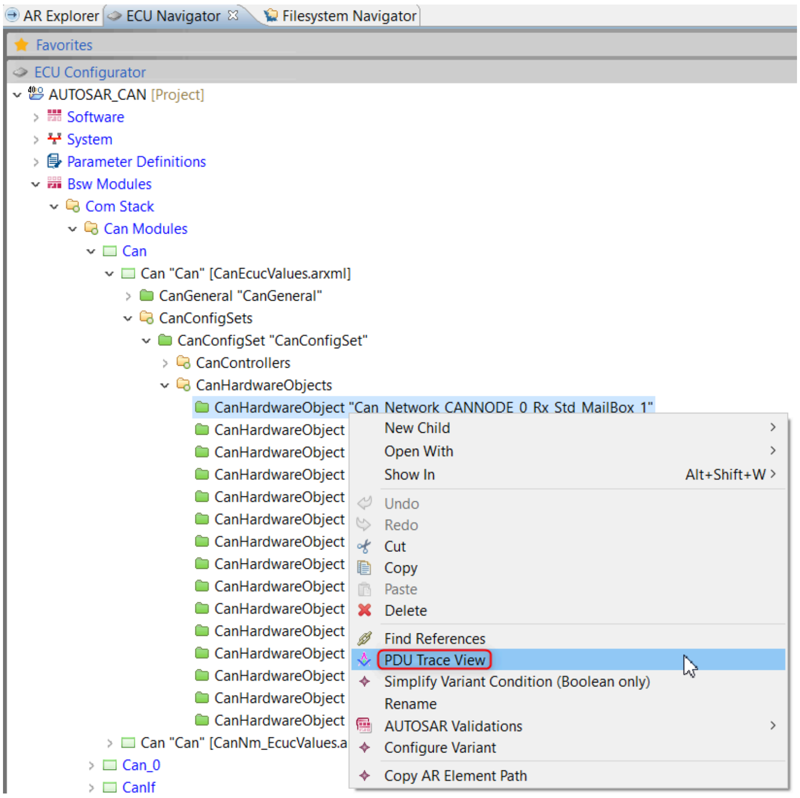

Figure 5.42.Invoking PDU Trace View

PDU Trace View supports the COM, TP, Nm, Ethernet, TCPIP, SoAd and Diagnostic modules, and the items displayed in the graphical view are colour-coded as follows:

- Yellow for Transmit
- Blue for Receive
- Sky Blue for the selection from which this view is opened (in this case PduRoutingPath)

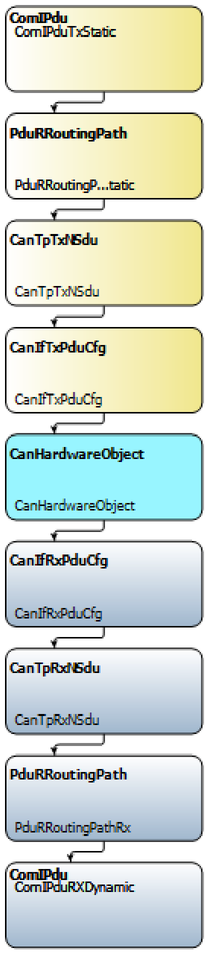

Figure 5.43.PDU Trace View

Note also the following points:

- Signal packing representation in the PDU can be visualized.
- Connection to the Application data, if possible, can be further processed and visualized.
- Double clicking on the other node opens the respective editor.

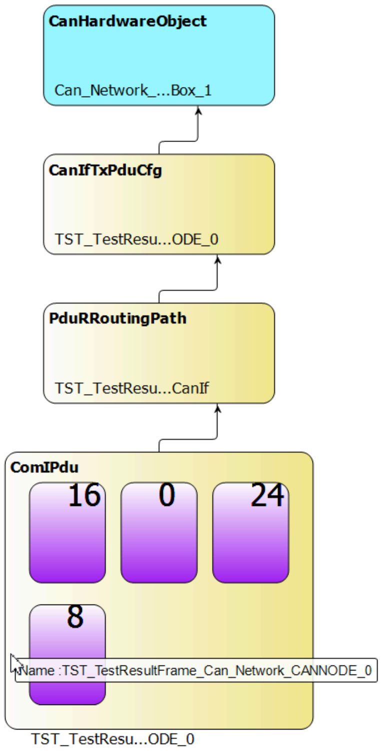

Figure 5.44.Signal packing

5.6.3 Channel Trace View

The Channel Trace View shows how ComMChannel handles different Communication Networks (E.g. CanSM, FrSM, CanIf, FrIf, Eth etc.).

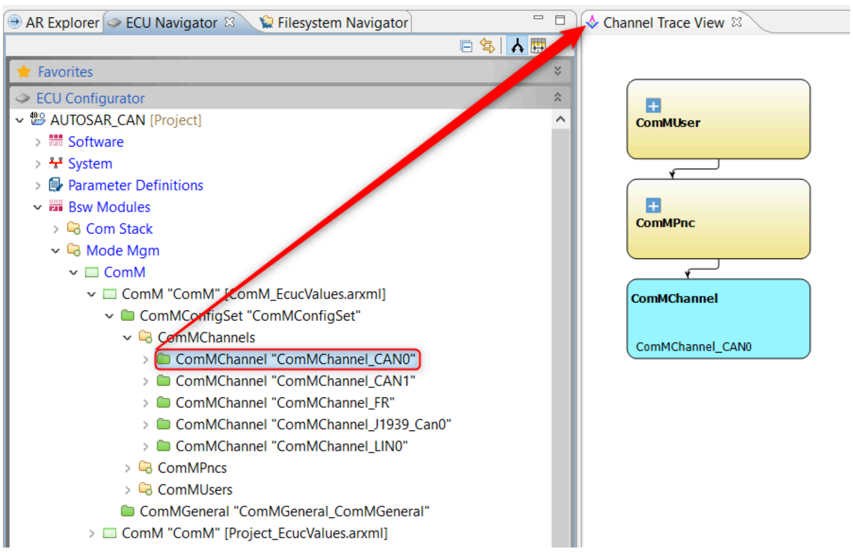

Figure 5.45.Channel Trace View

5.6.4 Diagnostic Trace View

The Diagnostic Trace View simplifies the analysis details about Diagnostic Events and shows more details about DTC, different Frame samplers, Enablement conditions and Storage of Records.

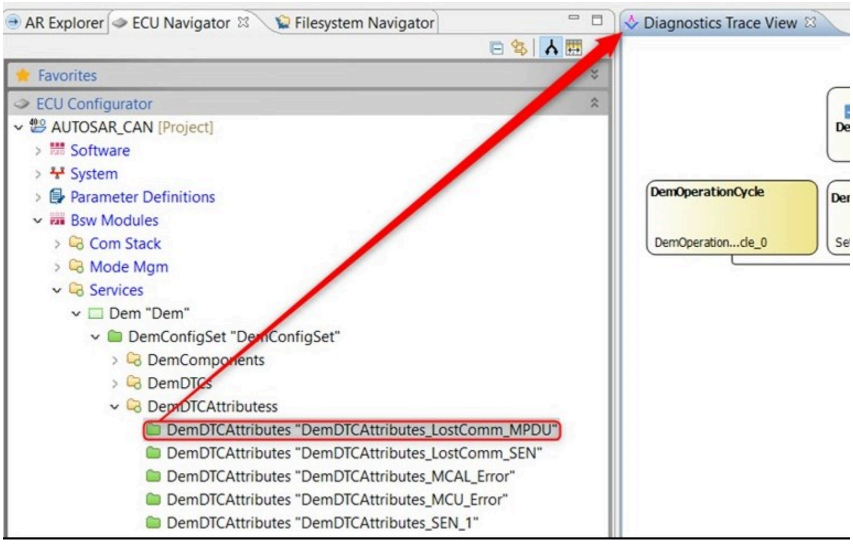

Figure 5.46.Invoking Diagnostic Trace View

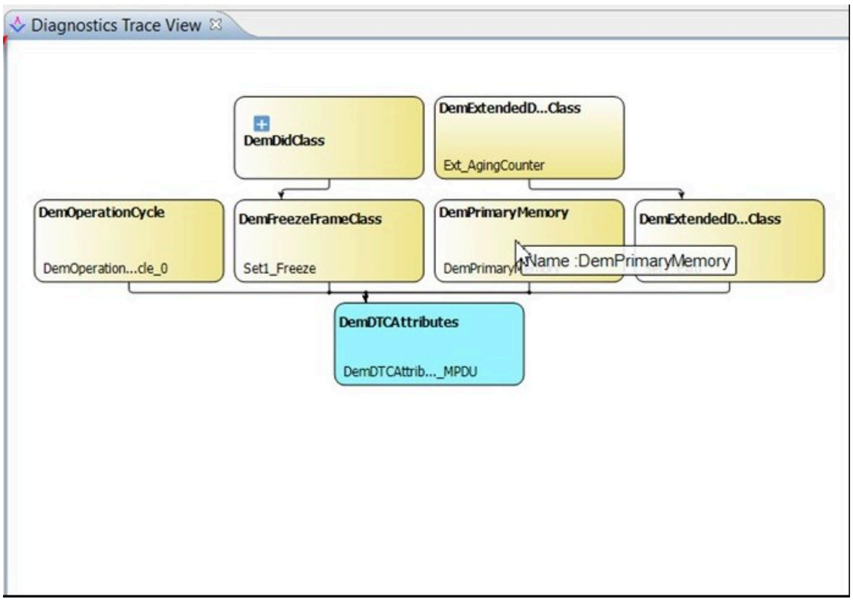

Figure 5.47.Diagnostic Trace View

5.6.5 OS Trace View

The OS Trace View simplifies the analysis details about the OS Application and shows more details about OS Tasks, ISRs, different OS Counters and Resources.

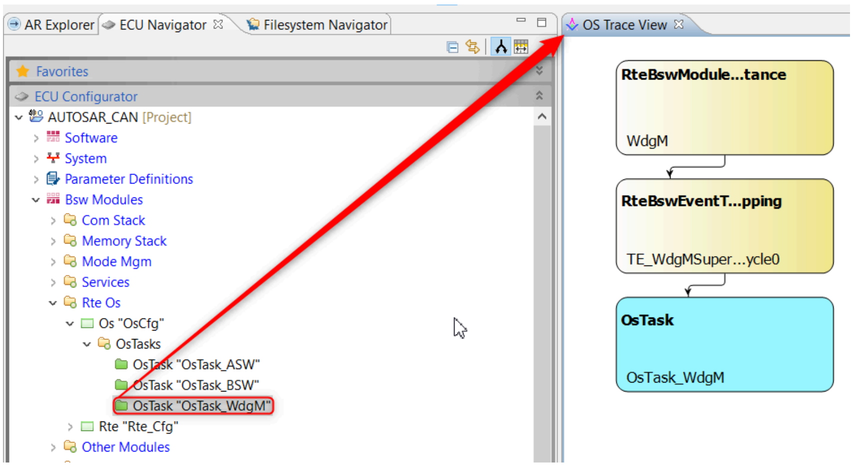

Figure 5.48.OS Trace View

5.6.6 Diagnostic Service View

The Diagnostics Service View simplifies the analysis details about DcmDsgServiceTable and shows more details about DcmDslProtocolRow, DcmService and DcmSubService.

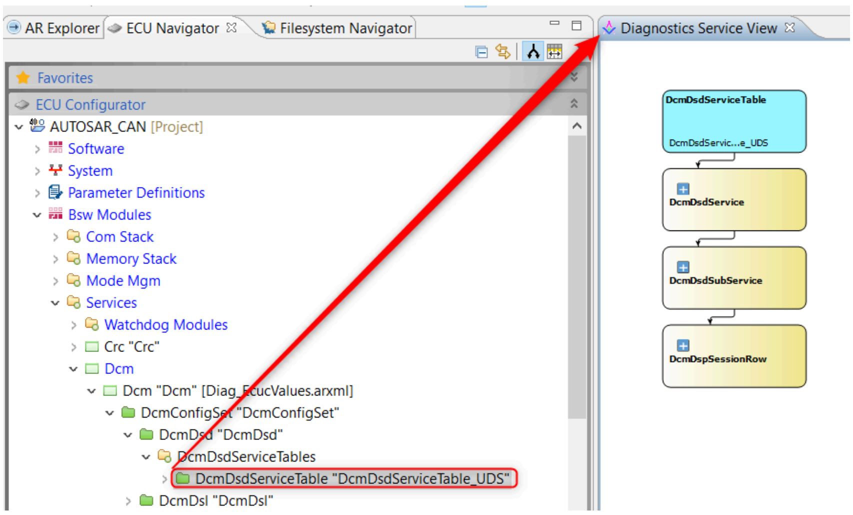

Figure 5.49.Diagnostic Service View

5.6.7 Ethernet Trace View

The Ethernet Trace View shows PDU Configured in Com Module and Ethernet controller.

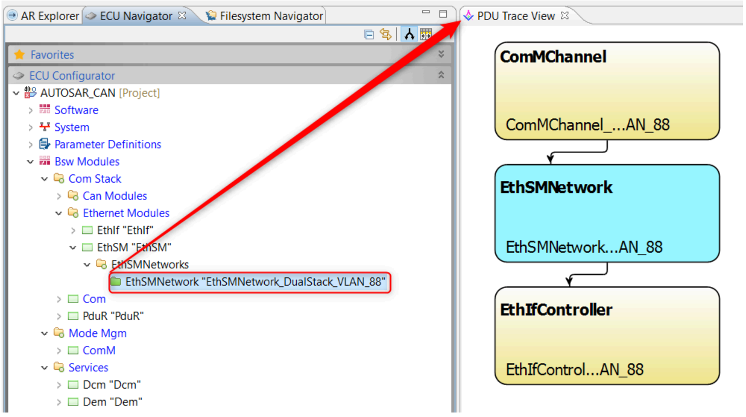

Figure 5.50.Ethernet Trace View

5.7 AUTOSAR Dashboard

5.7.1 Introduction

The AUTOSAR Dashboard allows you to visualize AUTOSAR use cases in one place, through a collection of charts and tables known as Dashboard Widgets. The Dashboard includes the following types of widgets:

- **Query:** Write a query to fetch the configured information from a Module or Container.
- **Network Architecture:** Visualize the network configurations in a tabular format. Selectable networks include CAN, Ethernet, Flexray, Flexray Timing Info, Flexray Global Info and LIN.
- **BSW Overview:** Visualize the configurations of modules as Pie or Bar charts.

5.7.2 Invocation

To open the AUTOSAR Dashboard go to **Window** › **Show View** › **Other** › **AUTOSAR Dashboard**.

Once inside the Dashboard, you can select/change the project in the Project selection dropdown.

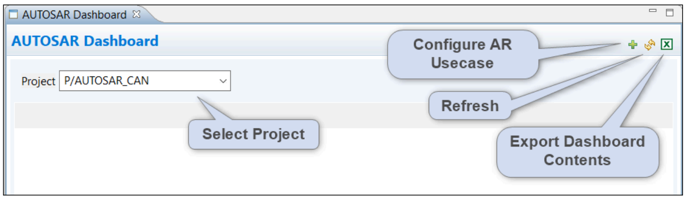

Figure 5.51.AUTOSAR Dashboard

5.7.3 Configuring Use Cases

To configure a use case, click on the **Configure AR Usecases** icon in the toolbar at the top of the Dashboard (as shown above in Figure 5.51).

In the “Configure AR Usecases” dialog, select the usecase that you wish to configure, and then click Next.

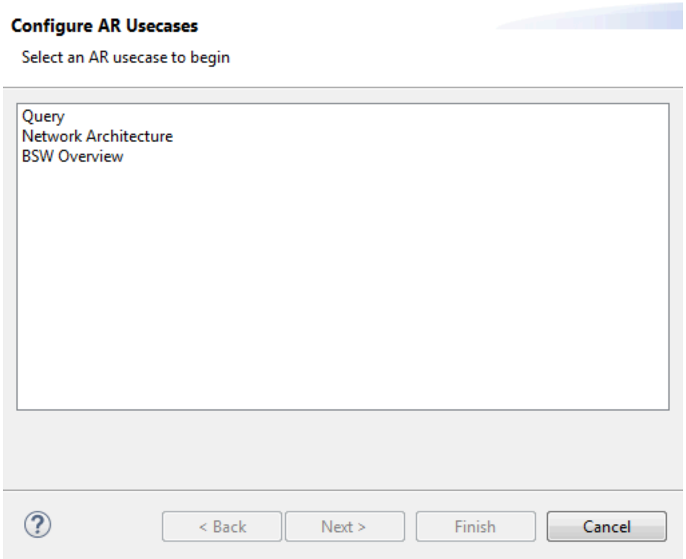

Figure 5.52.Configure AR Usecases

From our example above we've selected the **Query** usecase, and in the image below you can see that we can provide a Query Title and a Query. As an example, the following query will fetch the configured data from ComSignal containers:

```
FROM ComSignal WHERE ComBitSize > 0 SELECT shortName AS ShortName, ComBitSize, ComBitPosition, ComDataInvalidAction, ComErrorNotification, ComHandleId, ComInvalidNotification, ComSignalLength, ComFilter
```

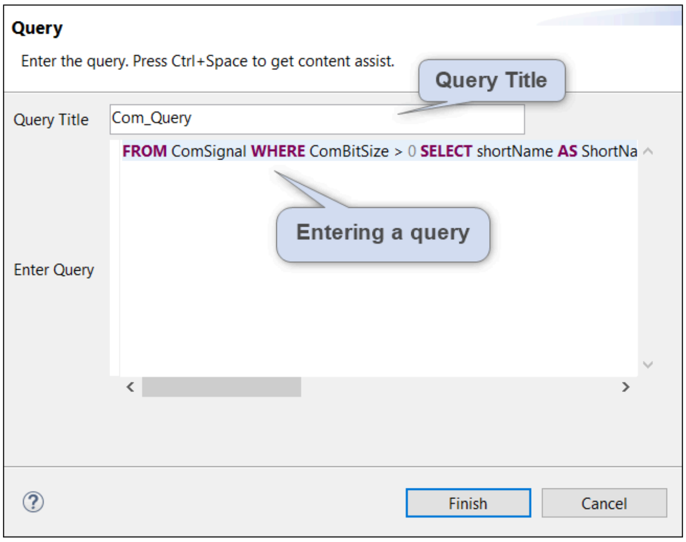

Figure 5.53.Configure Query Usecase

Here's an example of the generated query widget.

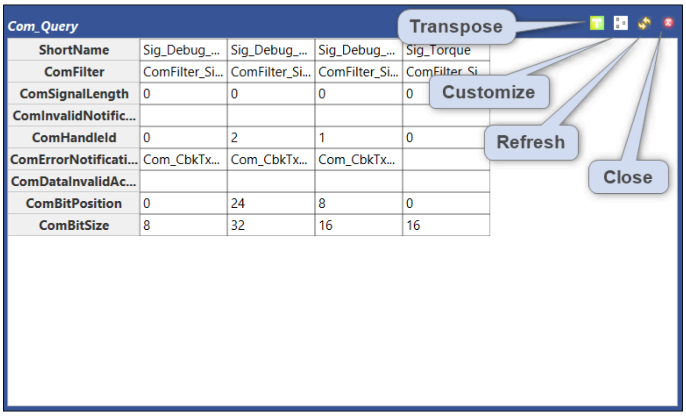

Figure 5.54.Generated Query Widget

If we select **Network Architecture** as the use case to configure, we can select the required **Network** and click **Finish**.

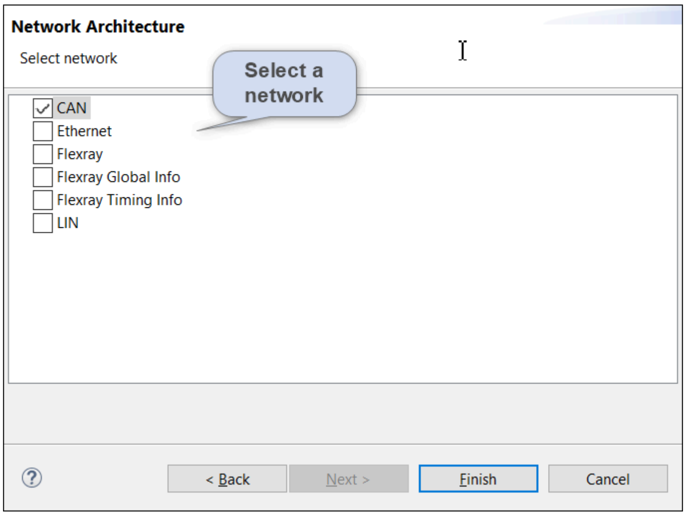

Figure 5.55.Configure Network Architecture Usecase

Here's an example of the generated CAN Network Architecture widget.

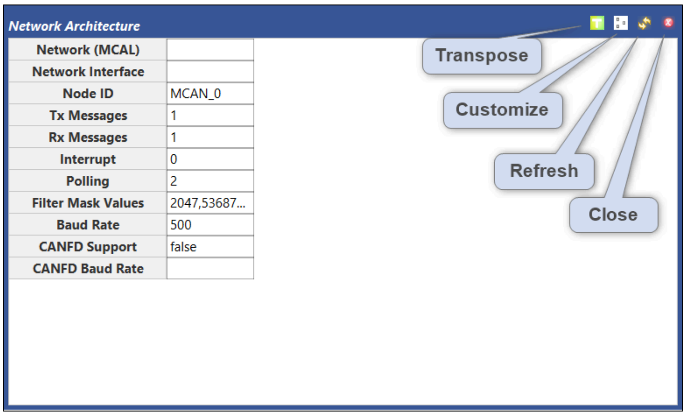

Figure 5.56.Generated CAN Network Architecture Widget

If we select **BSW overview** as the use case to configure, we can select the required Overview Type (Module or Stack), and then select the required Module(s) or Stack(s), or **Select All**. We can then choose the Visualization Type as either Pie Chart or Bar Chat.

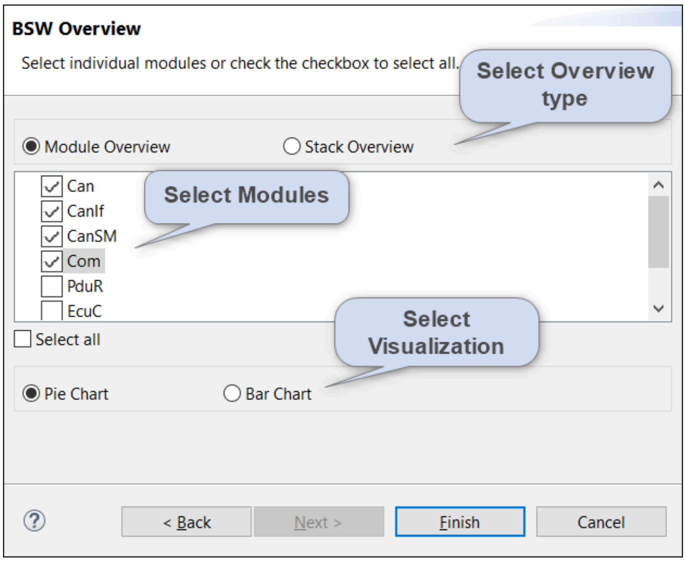

Figure 5.57.Configure BSW Usecase

Here's an example of the generated BSW Overview widget.

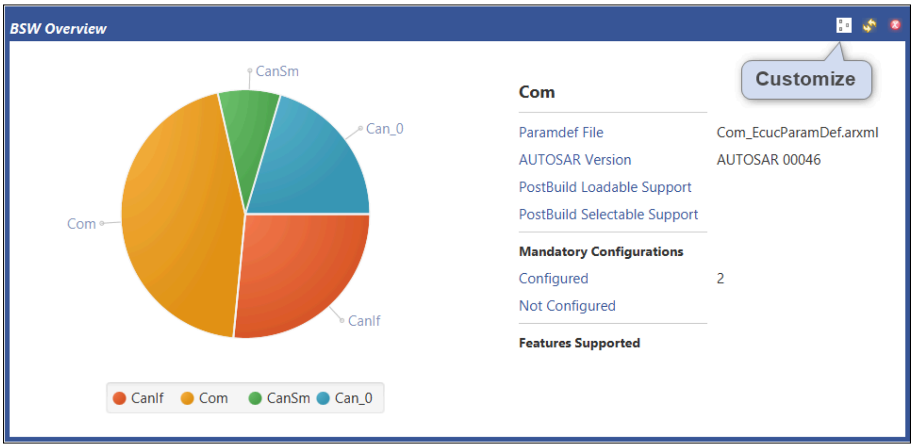

Figure 5.58.Generated BSW Overview Widget

5.7.4 Exporting Dashboard Contents

To export the contents of the configured use cases in the Dashboard, click the Export Dashboard Contents icon in the tool bar at the top of the Dashboard (see Figure 5.51).

In the “Export Dashboard Contents” dialog, set the File Name and Export Location for the export (or use the defaults), select usecase(s) and click **Export**. The selected contents of the Dashboard will be exported to the given location.

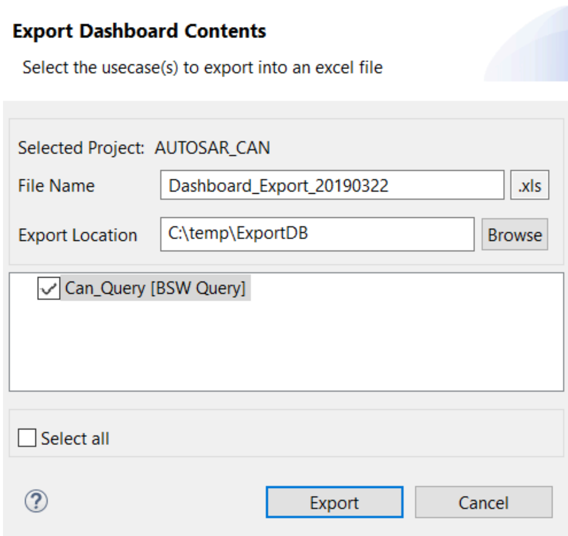

Figure 5.59.Export Dashboard Contents

5.8 Guided Configuration (for BSW Workflow)

5.8.1 Introduction

Guided Configuration (also known as Guided Workflow) provides you with a configuration path based on the workflow for configuring certain BSW software elements (e.g. Modules, Containers, Attributes configuration) and the validation of BSW modules, etc.

Once you've selected the use case, all the required workflow steps are displayed. If you execute them in the suggested order it will help you to complete the configuration more quickly.

Each of the steps listed in the workflow might display a dialog and ask you to provide input, or a step might open relevant editors/wizards through which the specific step can be completed. Some steps might open a sub-module with additional steps.

You can optionally create your own Guided Workflows, and they can be exported and imported if required.

5.8.2 Invocation

Guided Workflow can be invoked in one of two ways.

1. From the menu bar, select **Window** › **Show View** › **Other** › **ISOLAR-A** › **Workflow**.

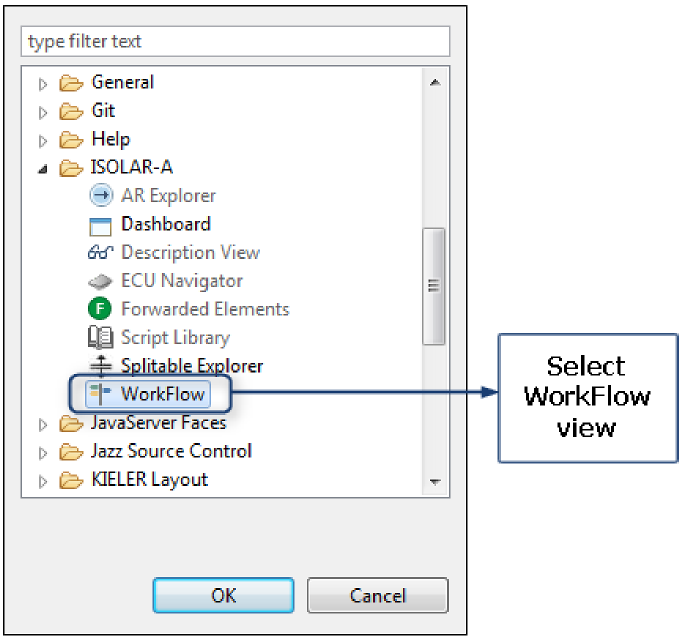

Figure 5.60.Invoking WorkFlow from the menu bar

2. Or, in the "AR Explorer", right-click on the project, then select **ISOLAR-A** ›

WorkFlow.

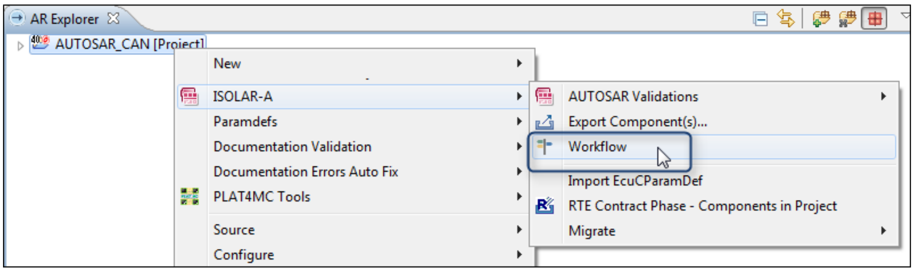

Figure 5.61.Invoking WorkFlow from AR Explorer

5.8.3 Workflow View

Here is an example of the Workflow View, with the key features highlighted.

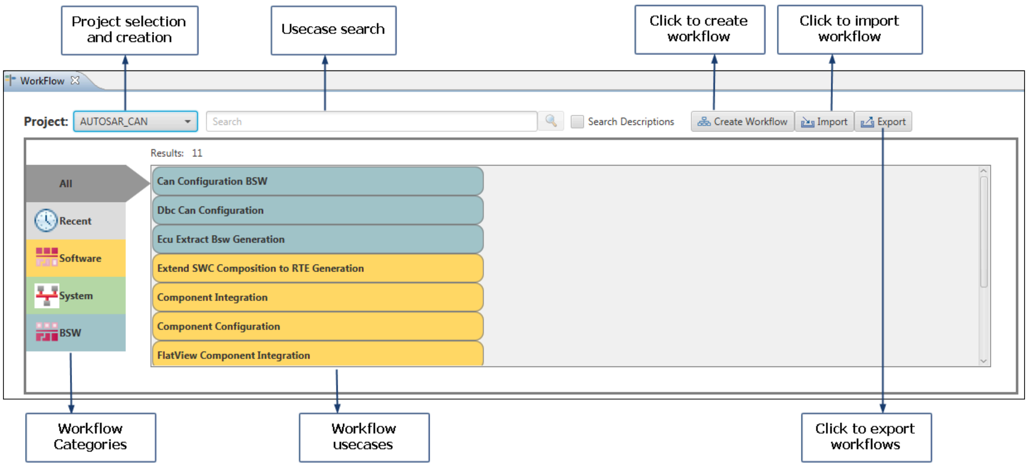

Figure 5.62.Workflow View

5.8.4 Creating a Custom Workflow

To create a Custom Workflow, click on the **Create Workflow** button, select the required BSW category in the left hand pane, then drag and drop the required action(s).

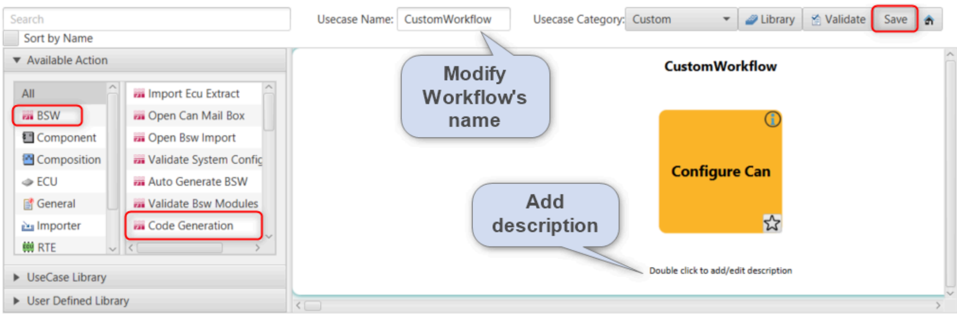

Figure 5.63.Create a Custom Workflow

Click **Save** to save your updates, and click the **Home** button to return to the Workflow Dashboard.

5.8.5 Adding Conditions to the Workflow

Workflow Conditions can be added to the BSW steps/modules in a custom workflow.

First, configure a custom workflow, and then right-click on any step/module within the workflow and select "Add Condition".

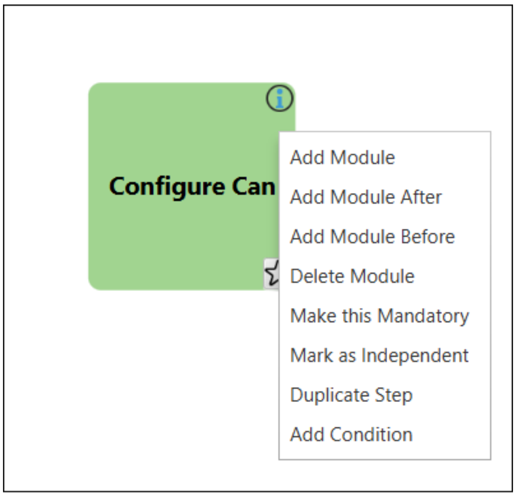

Figure 5.64.Add Condition

In this first example we're adding an Ecuc Parameter Condition. We set the parameter, operator and value for the condition (e.g. CanBusOffProcessing == POLLING), and clicked **Add**.

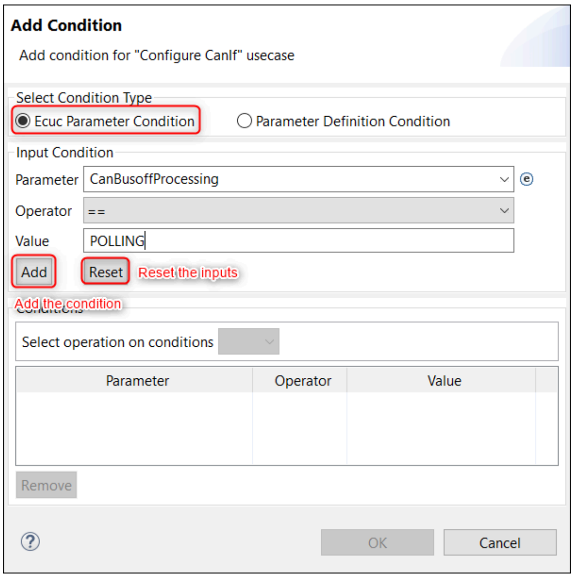

Figure 5.65.Add EcucParameter Condition

And in this example we're adding a Parameter Definition Condition. We select the parameter from the parameter combo and click Add.

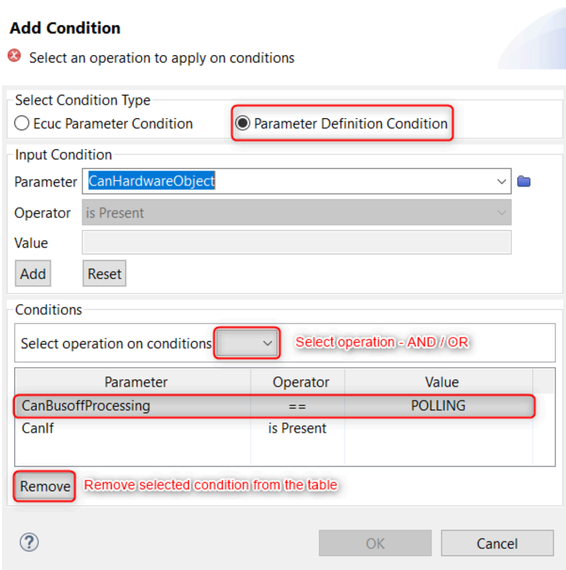

Figure 5.66.Add Parameter Definition Condition

If you need to remove a condition, select it from the Conditions table and click **Remove**.

If you're setting multiple conditions, select an operation (AND/OR) from the operation selection combo.

5.8.6 Parameter Definition Validations

Parameter Definition Validations are performed during the launch of a custom workflow for the BSW steps/modules with an associated paramdef.

If the paramdef corresponding to a step or module is not configured in the selected project, this is highlighted as an error, as shown here.

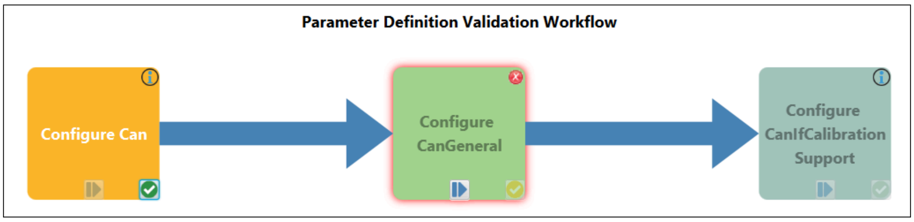

Figure 5.67.Parameter Definition Validation

5.8.7 Creation of non-existing Modules/Containers

On executing a step/module in a custom workflow, if the Module or Container (value) is not created, the workflow allows for the auto-creation of the missing Module or Container.

Execute a step/module after launching the custom workflow (e.g. Can), as shown here.

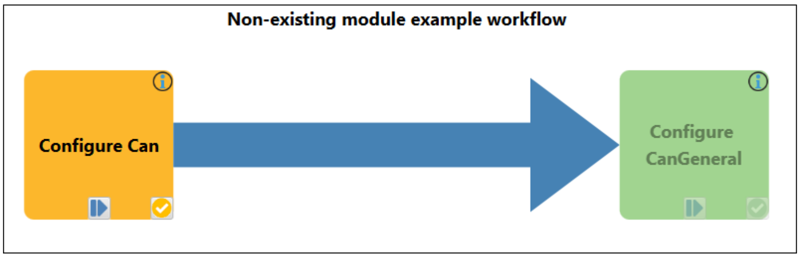

Figure 5.68.Auto-create a non-existing Module/Container

A parent selection dialog will then be prompted (see next page) in which you select the immediate parent of the Can module.

Select the ARPackage from the dialog under which the Can module (value) will be created.

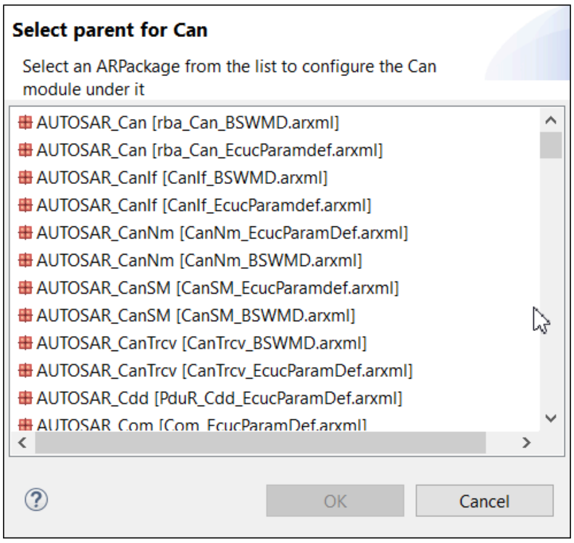

Figure 5.69.Parent selection

The created Can module will open in the BSW Editor.

5.9 M2M Framework (System to BSW)

5.9.1 Introduction

The M2M Framework provides solutions for many Model to Model transformation use cases. Currently the System-to-BSW (A2B Transformer) use case is implemented using the M2M Framework.

5.9.2 Feature Summary

Features of the M2M Framework:

- A Mapping Editor is provided to create mappings and to enhance the existing mappings provided by AUTOSAR.
- ECU Feature Selection.
- Default values to mappings are provided.
- Condition based mapping execution.

5.9.3 Invocation

The M2M Editor is accessed by right-clicking on a project, then selecting **Open With** › **Model Mapping Editor**.

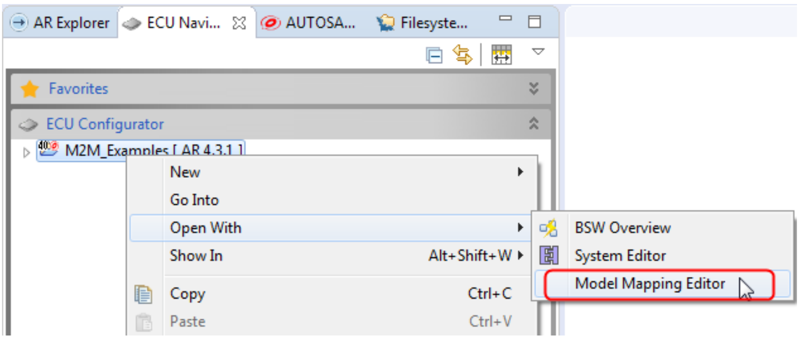

Figure 5.70.Invoking the Model Mapping Editor

5.9.4 Use Cases

5.9.4.1 Use Case 1: System to BSW (A2B Mapping)

1. The **Mapping Editor** is used to create new mappings, edit/enhance the existing mappings provided by AUTOSAR, and add/edit parameter mapping.
2. The **Mapping Summary** is used to display use cases, show their different mappings, and show conditions and default values for attributes.

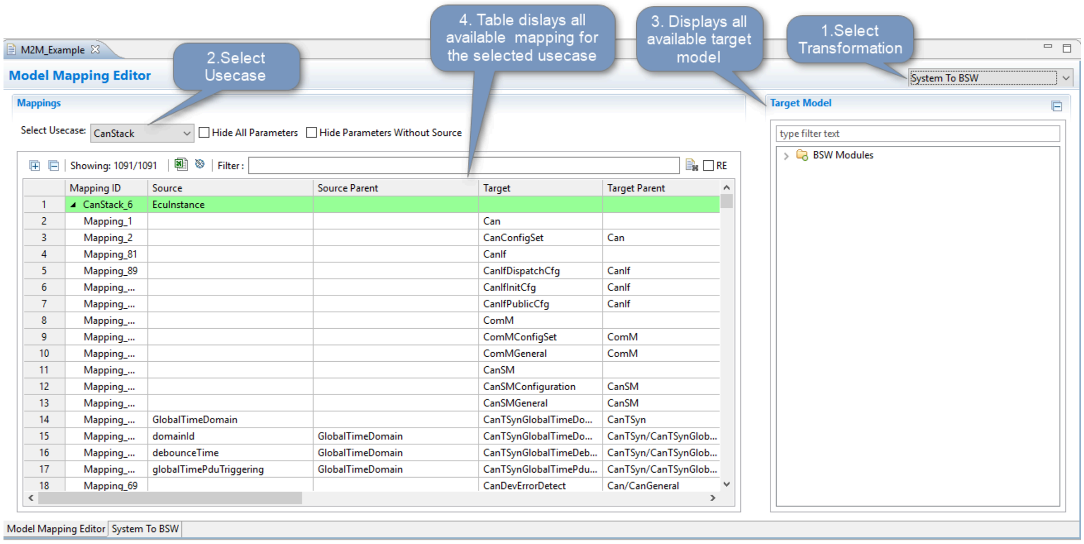

Figure 5.71.System to BSW - A2B Mapping

5.9.4.2 Use Case 2: System to BSW (BSW Configuration Generation)

1. Import the M2M_Example project from *‹product_install_location›/ExamplesForCustomization05_examples_for_m2mexample_projectM2M_Example.zip*
2. Right click on the project, select **Open With** › **Model Mapping Editor**, and select the “System to BSW” tab as shown below.
3. Select “Can Stack” and click **Invoke System to BSW** to generate the BSW configurations.

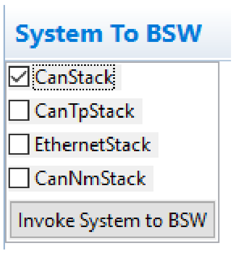

Figure 5.72.System to BSW - BSW Configuration generation

5.10 Conf-Gen Enhancer

5.10.1 Introduction

The Conf-Gen Enhancer can be used to apply default values for BSW parameters to the generated configuration (from the Synchronize Editor). This involves configuring default values for a module in a mapping file in the Mapping Editor, and then invoking the Conf-Gen, during which the generated module will have the defaults applied.

5.10.2 Configuring Default Values

The first step is to configure the default values for the BSW parameters for each module. From the ECU Navigator, right-click on the project, then select **Open With** › **Model Mapping Editor**.

1. Select the Transformation as “Conf-Gen Enhancer”.
2. Select the Mapping Usecase to match the module for which you are configuring the defaults.
3. Drag and drop the parameters from the tree on the right into the Mappings table.
4. Configure the default value for each parameter in the Default Value column.

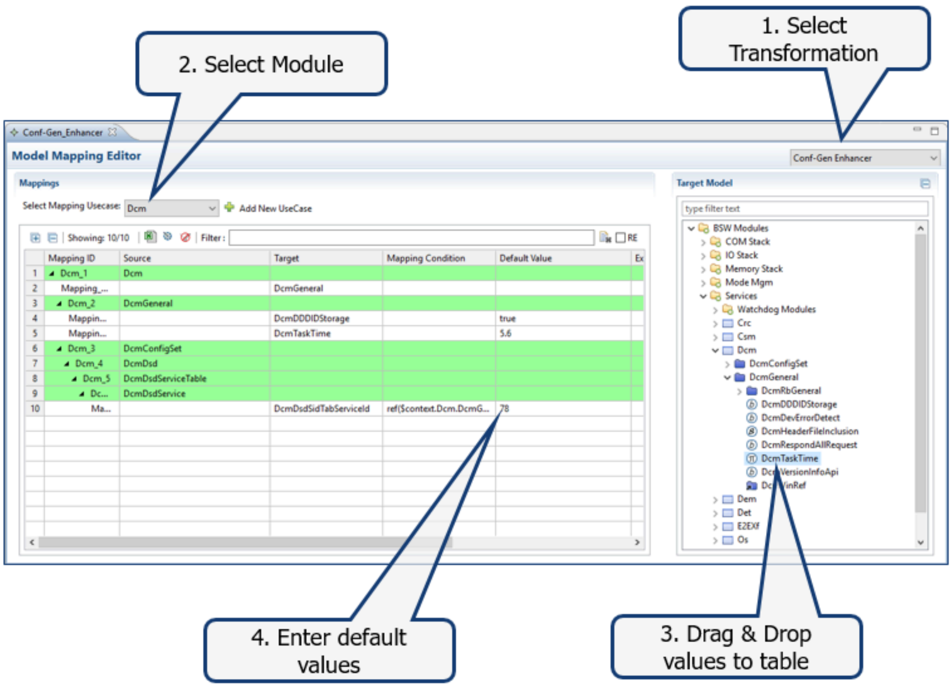

Figure 5.73.Configure default values

When you’re finished, click **Save**, and the default values will be written to a *‹module_name›.mapping* file.

5.10.3 Invocation

With the default values now setup, they're ready to be used with the Conf-Gen Enhancer.

1. Place the mapping file for the required module(s) in the Conf-Gen_Enhancer folder in the project.

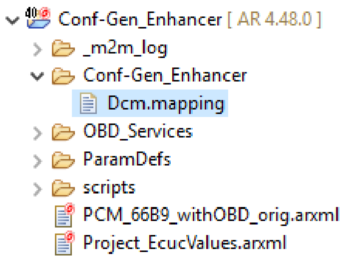

Figure 5.74.Mapping file for the Conf-Gen Enhancer

2. Right-click on the project, and then open the System Editor by selecting **Open With** > **System Editor**.
3. Select the script that you want to execute, then click **Synchronize**.
4. If a mapping file is present in the project for the generated module, the default values for the configured parameter/containers will be applied to that module.

5.11 Configuration Logger

5.11.1 Introduction

The Configuration Logger logs all the configuration changes that have been made in the workspace, together with details of the UI actions that led to each change. The following types of changes are logged:

- Addition/Deletion of elements
- Modification of an element's properties
- Addition/Deletion of files
- Addition/Deletion of a project

5.11.2 Invocation

The Configuration Logger should be enabled by default, but if it is not visible, it can be accessed via **Window** › **Show View** › **Other...** and then selecting **ISOLAR-A** › **Configuration Logger**.

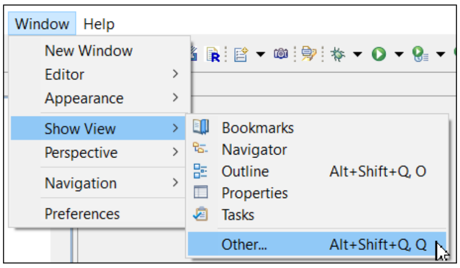

Figure 5.75.Invoking Configuration Logger

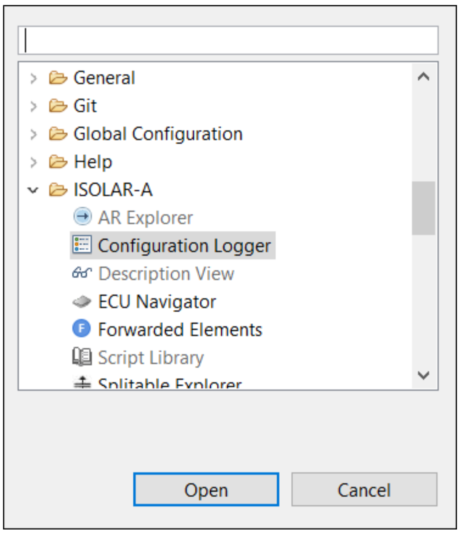

Figure 5.76.Invoking Configuration Logger

5.11.3 Options

The image below summarizes the key functionality of the Configuration Logger, which includes the following:

Project Filter: By default, changes related to all projects are shown, but you can use the Project Filter to see only the changes for a specific project.

Start/Stop Logging: Allows you to Start/Stop the logging.

Clear and Backup Configuration Log: Allows you to clear the log and optionally take a backup of the configuration log file.

Undo: The undo icon is displayed next to any action that can be undone. When the changes related to the selected action have been undone, the undo button is then disabled.

Roll back: The roll back icon is displayed next to any action that can be rolled back (note that not all actions can be rolled back). All the eligible actions up to the selected action are rolled back, and once the operation is completed, the roll back icon that was displayed next to each rolled back action will be disabled.

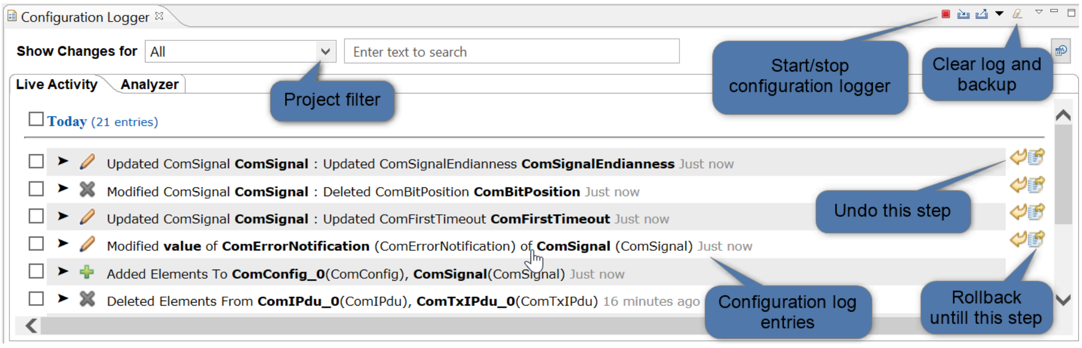

Figure 5.77.Configuration Logger

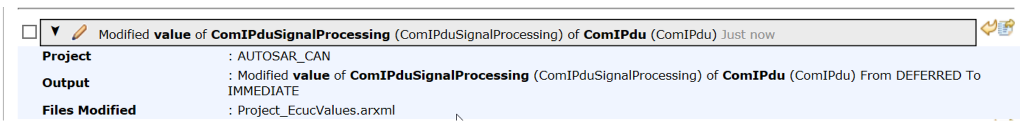

Figure 5.78.Configuration Logger description

5.12 Synchronize with Conf-Gen Bundle Execution

5.12.1 Introduction

BSW configuration can be generated using scripts or using the RTA_BSW bundle, which contains scripts, defaults and mapping files that are not part of the project and which are used for BSW generation.

5.12.2 Invocation

BSW configuration generation can be executed in the BSW page in the System editor.

Right click on the Project in which you want to run the Conf-Gen, then select **Open With** › **System Editor**.

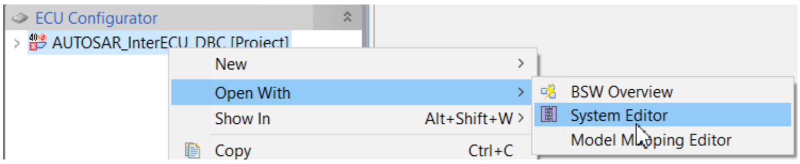

Figure 5.79.Opening the project in System Editor

The image below summarizes the main features of the BSW page in the System Editor.

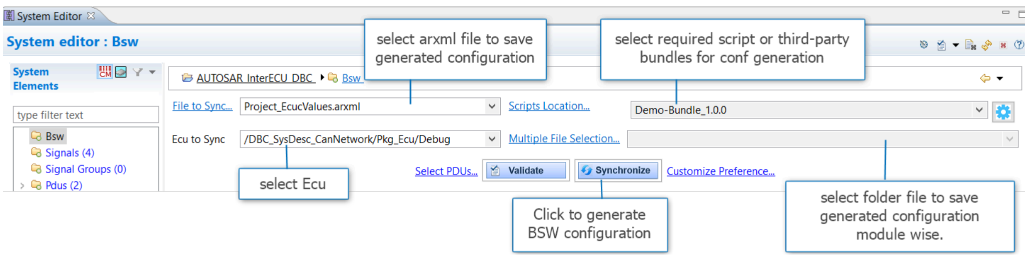

Figure 5.80.System Editor (BSW page)

5.12.3 Usecase

The following steps would be required to do a BSW generation using an RTA_BSW bundle.

1. Import a project that does not contain scripts.
2. Install the RTA_BSW bundle using an update site.
3. Right click on the project, then select **Open With** › **System Editor**.
4. In the Bsw page, select the required bundle in Scripts Location
5. Click the Synchronize button.

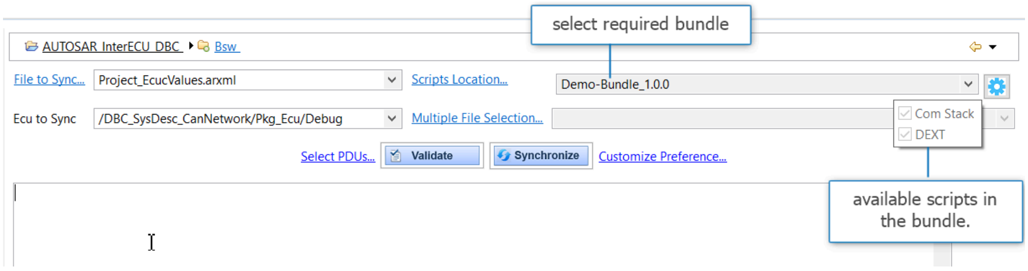

Figure 5.81.Select bundle

The BSW configuration will be generated, and all default values will be applied.

5.13 BswM Auto Configuration

5.13.1 Introduction

The BswM Auto Configuration dialog helps you to automatically configure generic /common BswM use cases just by selecting the required use case and the required inputs.

5.13.2 Invocation

The BswM Auto Configuration dialog can be opened from the ECU Navigator by right-clicking on Bsw Modules, then selecting **Create Mode Mgm** › **Auto Configure BswM**.

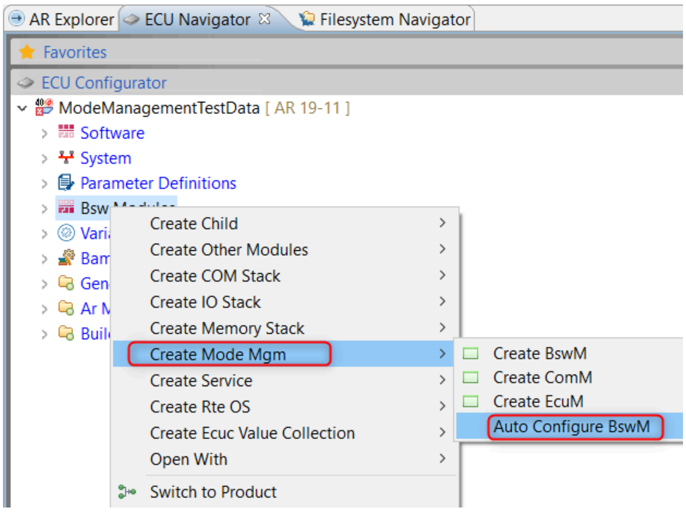

Figure 5.82.Invoking BswM Auto Configuration

5.13.3 Overview

The BswM Auto Configuration dialog shows the BswM use cases that are supported.

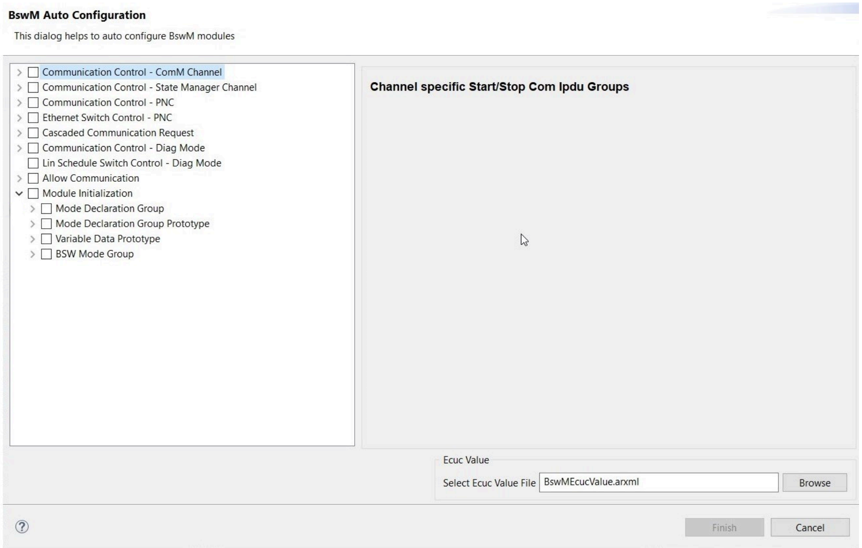

Figure 5.83.BswM Auto Configuration dialog

5.13.4 Use case input selection

Follow these steps to select the required input for a use case.

1. In the left-hand pane, select and then expand the use case that you wish to run. The possible elements for the configuration of BswM will be shown.
2. In the right-hand pane, all the required input for the selected use case will be displayed. Select all the inputs for the BswM configuration.
3. In the “Select Ecuc Value File” box, select/enter the name of the Ecuc Value file.
4. When you click Finish, the BswM module will be auto generated for the selected use cases.

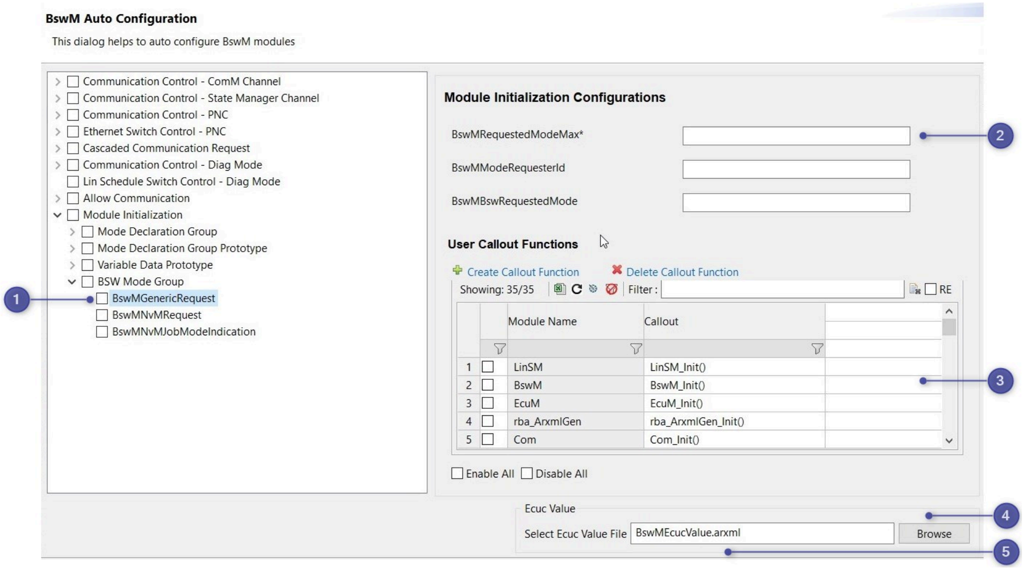

Figure 5.84.BswM Auto Configuration - Input selection

No.	Description
1	Use case selection under Module Initialization
2	Required inputs to auto configure BswM module with arbitration (mode request source, condition, rule and expressions)
3	Action lists and action item
4	Browse ECUC-value file from the project
5	Edit the file name

The following elements will be generated in the EcucValue file:

- Mode request port
- Mode condition
- Logical expressions
- Rules
- Actions
- Actions list

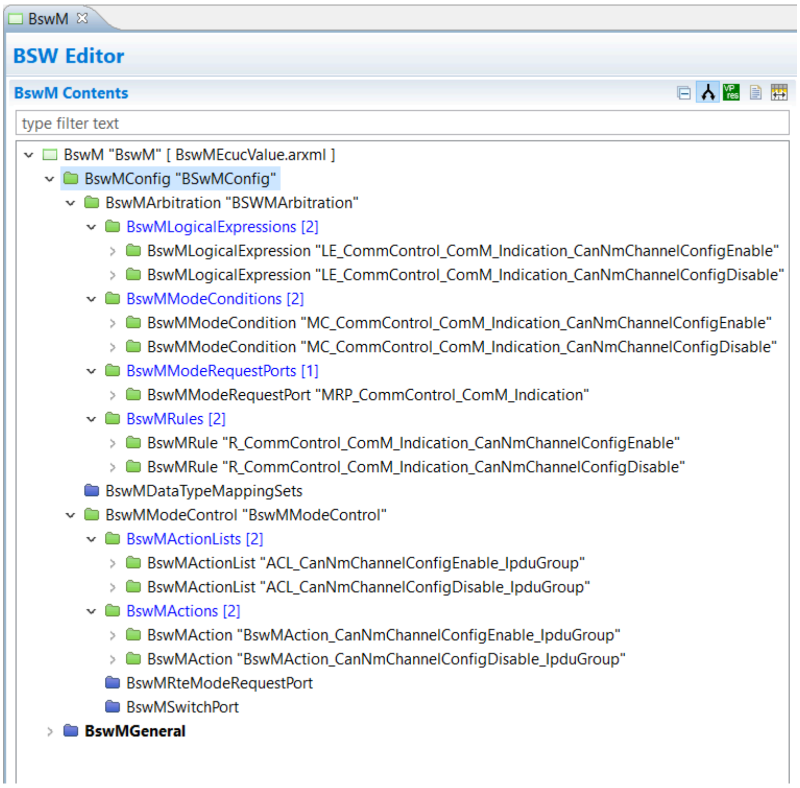

Figure 5.85.Auto generated BswM module

6 ETAS Contact Addresses

6.1 Technical Support

Technical support is available to all users with a valid support contract. If you do not have a valid support contract, please contact your regional sales office.

- https://www.etas.com/en/contact.php

The best way to get technical support is using the online support portal RTA Hotline:

- https://rtahotline.etas.com/

It is helpful if you can provide technical support with the following information (where relevant):

- Your support contract number
- The version of the ETAS tools you are using
- The version of the compiler tool chain you are using
- The command line (or reproduction of steps) that result in an error message
- The error messages or return codes you received (if any)
- Your RTA-CAR Project or if not possible the arxml file(s) containing the error(s).
- The *Diagnostic.dmp* if it was generated

6.2 ETAS HQ

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